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Patidar et al.

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(54) **METHOD OF TESTING AN INTEGRATED CIRCUIT AND TESTING SYSTEM**

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(58) **Field of Classification Search**
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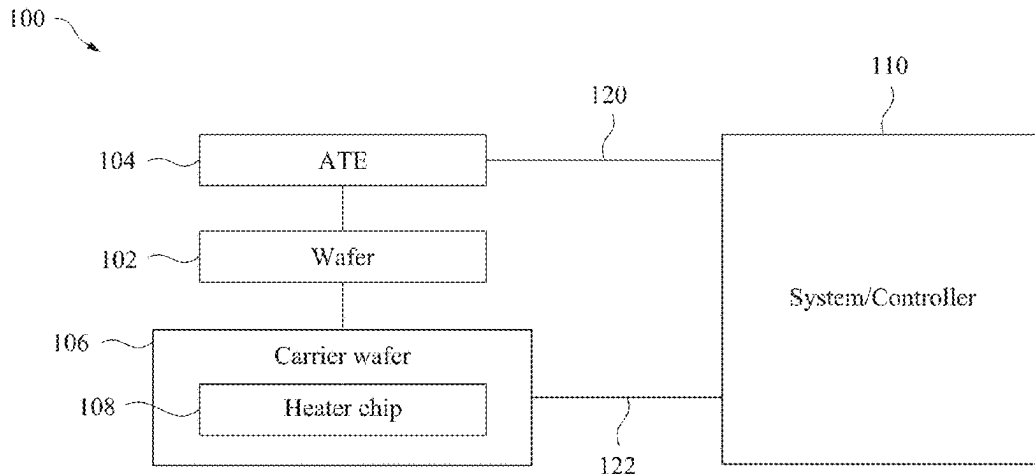
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(57) **ABSTRACT**
A method of testing an integrated circuit on a test circuit board includes performing, by a processor, a simulation of a first heat distribution throughout an integrated circuit design, and simultaneously performing a burn-in test of the integrated circuit and an automated test of the integrated circuit. The burn-in test has a minimum burn-in temperature of the integrated circuit or a burn-in heat distribution across the integrated circuit that includes a set of circuit blocks or
(Continued)



a first set of heaters. The integrated circuit design corresponding to the integrated circuit. The performing the simulation includes determining a heat signature of the integrated circuit design from configured power information or location information for each circuit block of the set of circuit blocks or each heater of the set of heaters included in the integrated circuit design. The heat signature includes heat values distributed throughout the integrated circuit design.

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G06F 119/08 (2020.01)

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G06F 30/398; G06F 2119/08; G06F
2119/06; G06F 30/39

See application file for complete search history.

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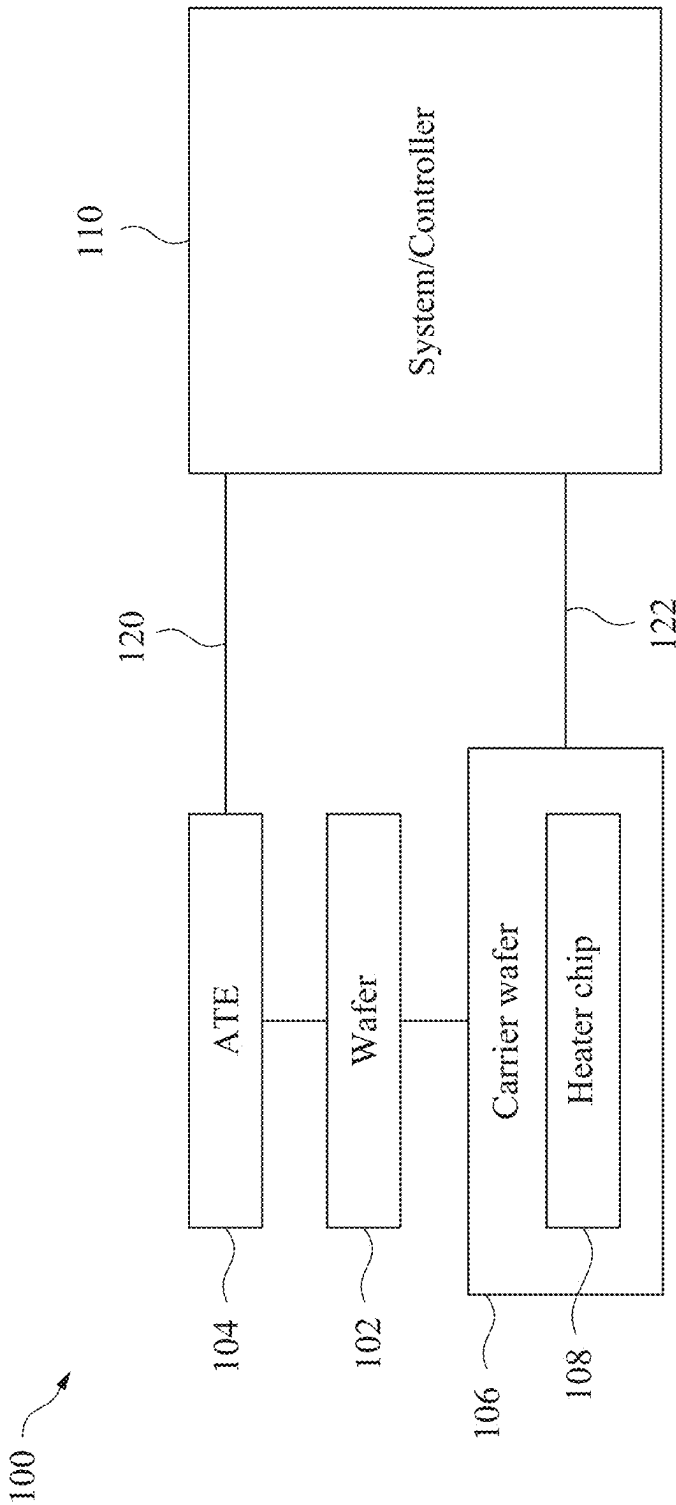


FIG. 1

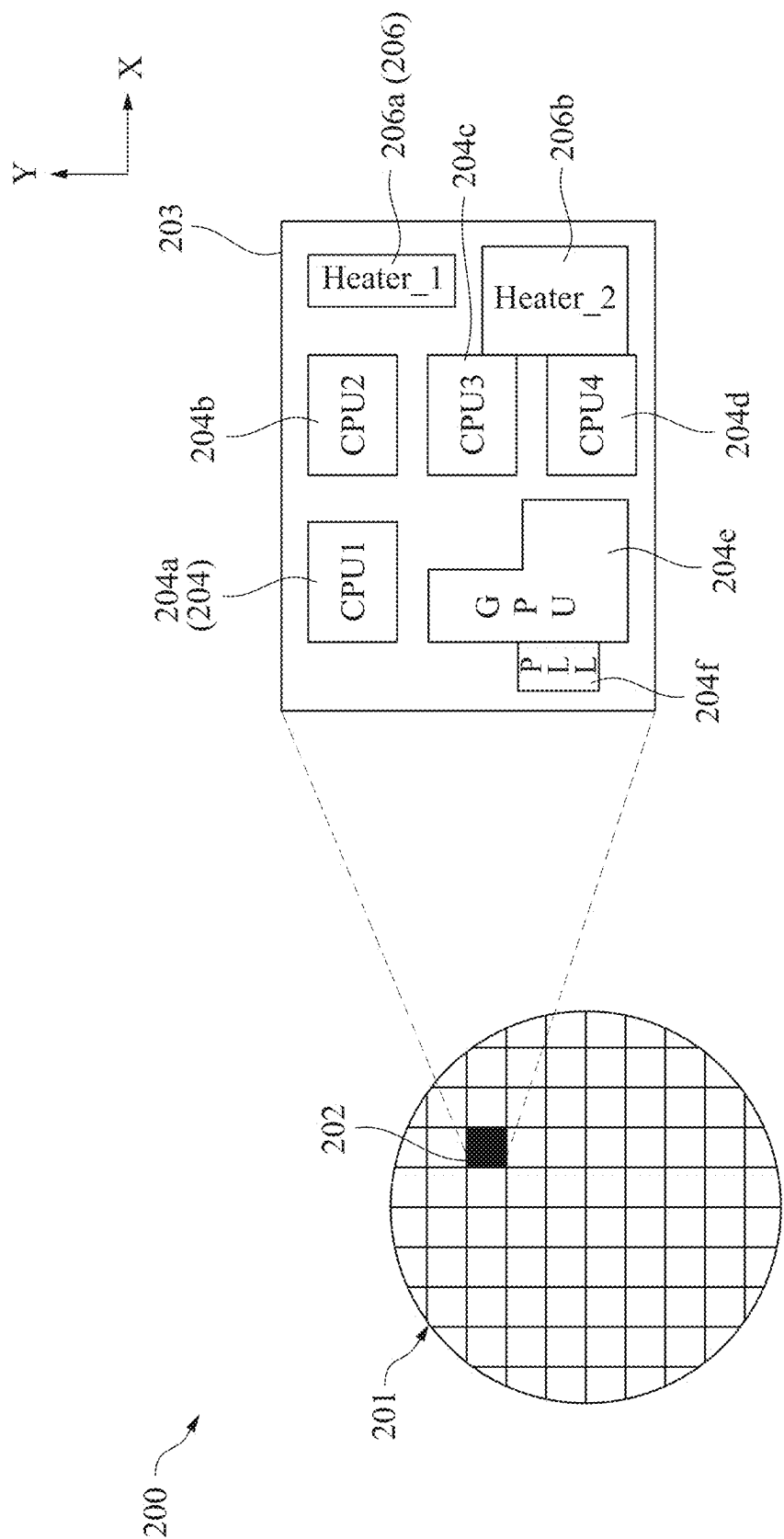


FIG. 2

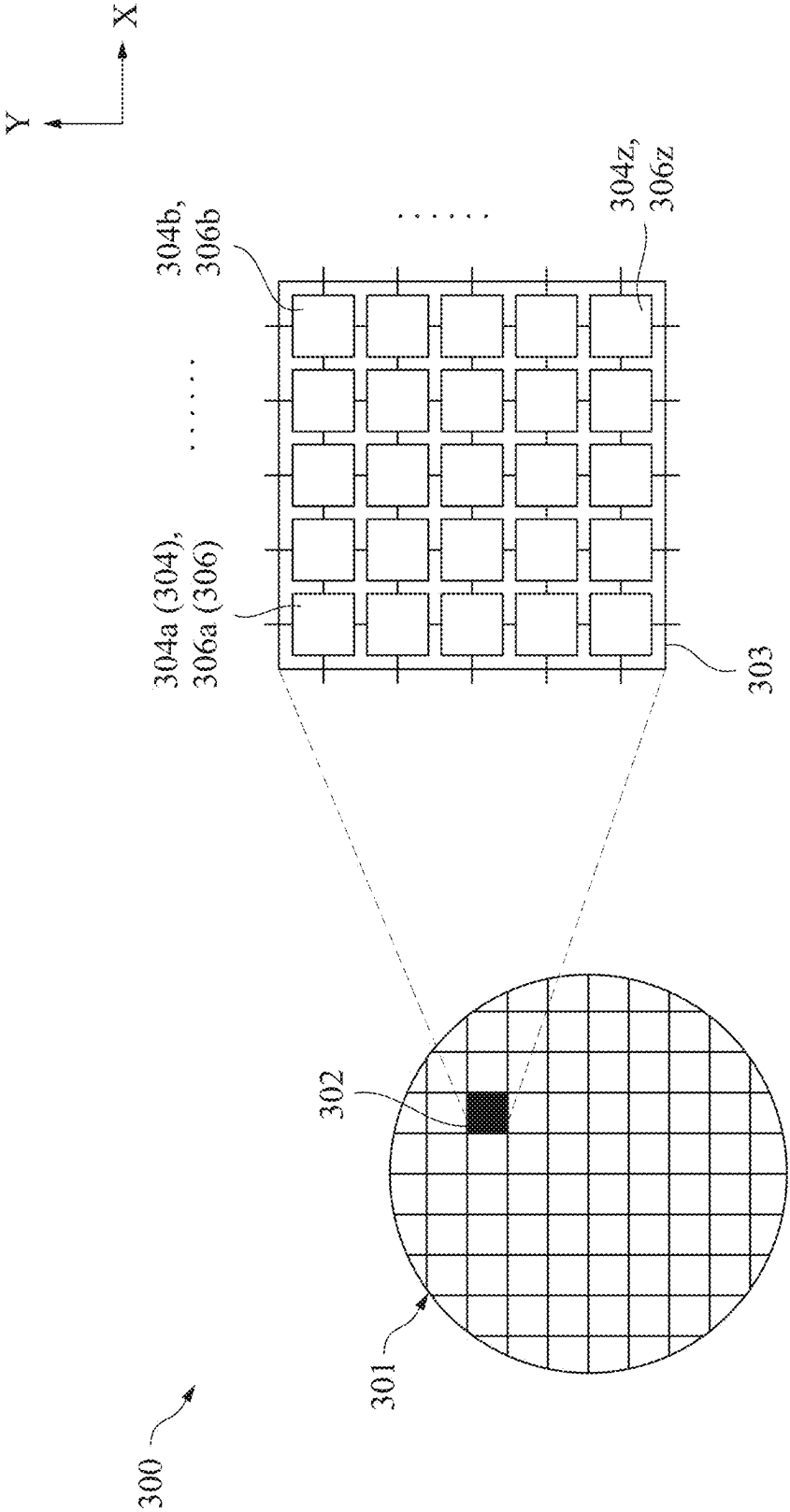


FIG. 3

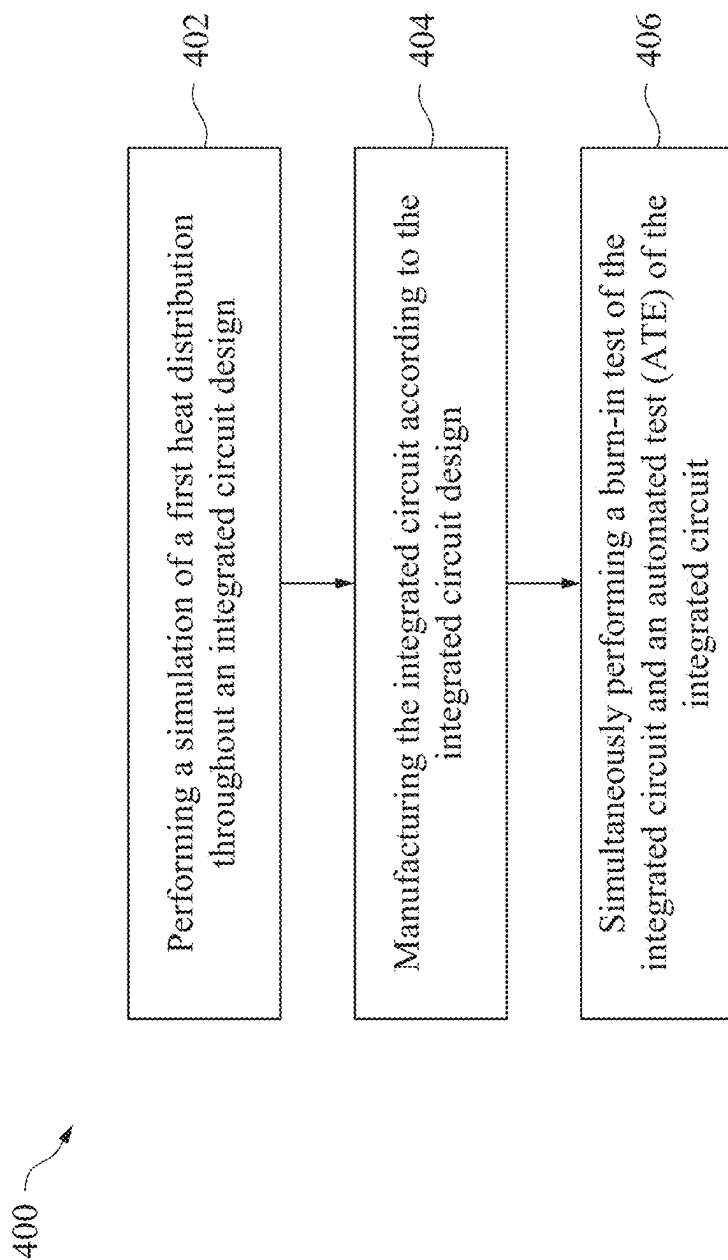


FIG. 4

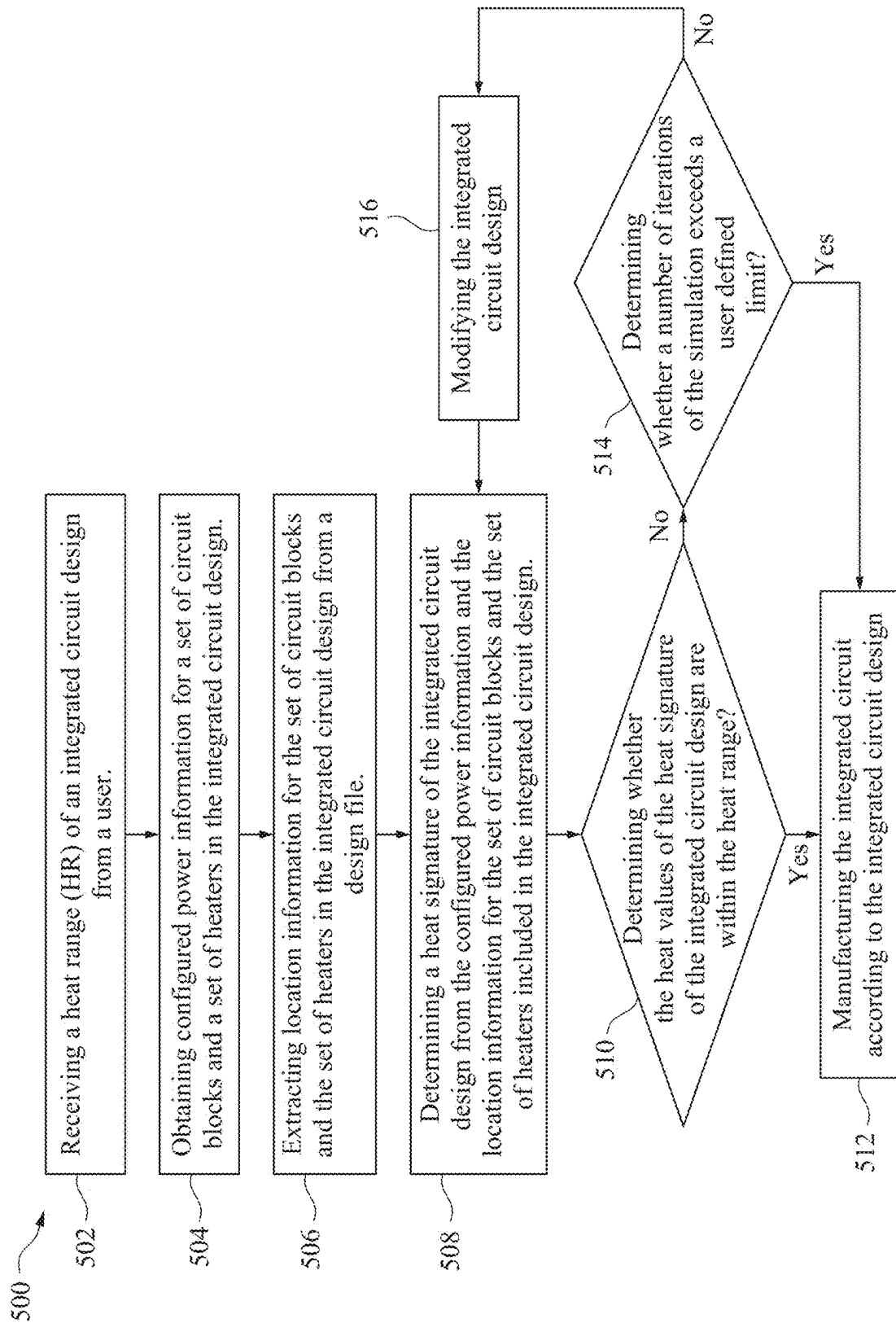


FIG. 5

600

Column 1		Column 2		Column 3
IC Design Elements		Max Power	Configurable Power	
602a	CPU	10W	606a	
602b	GPU	8W	606b	
602c	Heater_1	2W	606c	
602d	Heater_2	5W	606d	
602e	PLL	0.5W	606e	
		604e 604d 604c 604b 604a		
		606		

FIG. 6

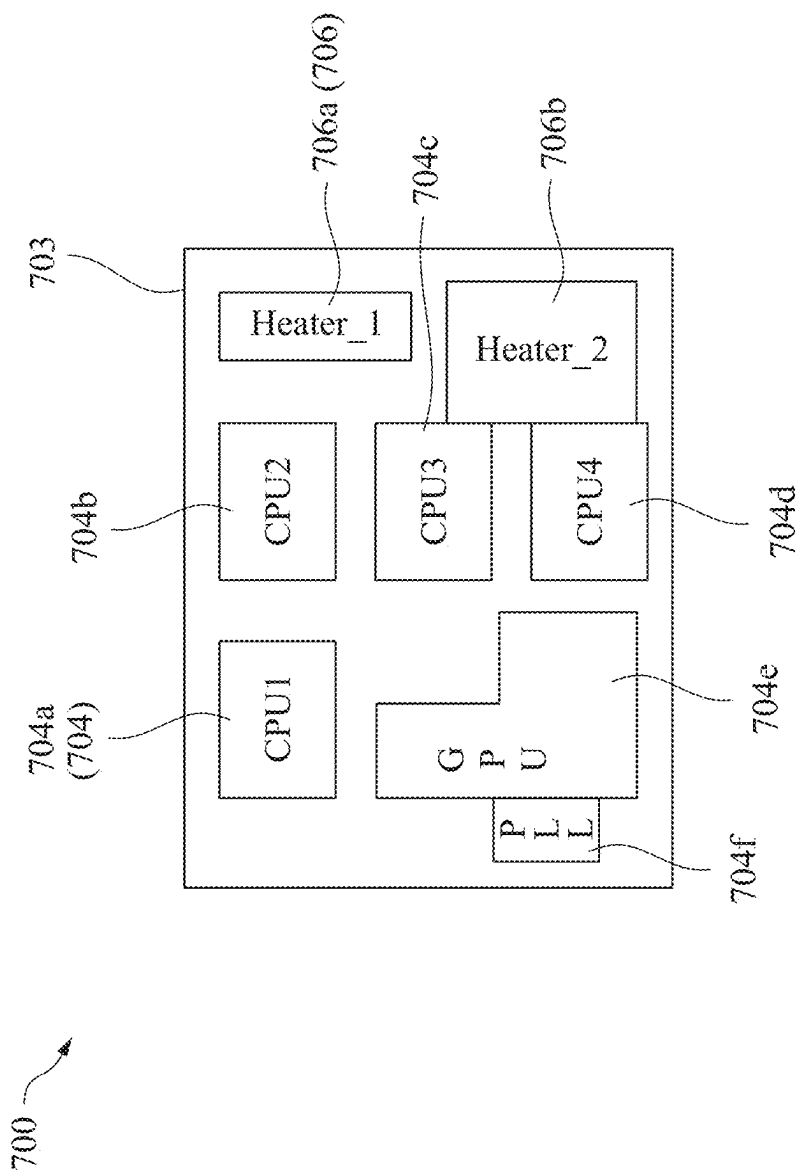


FIG. 7

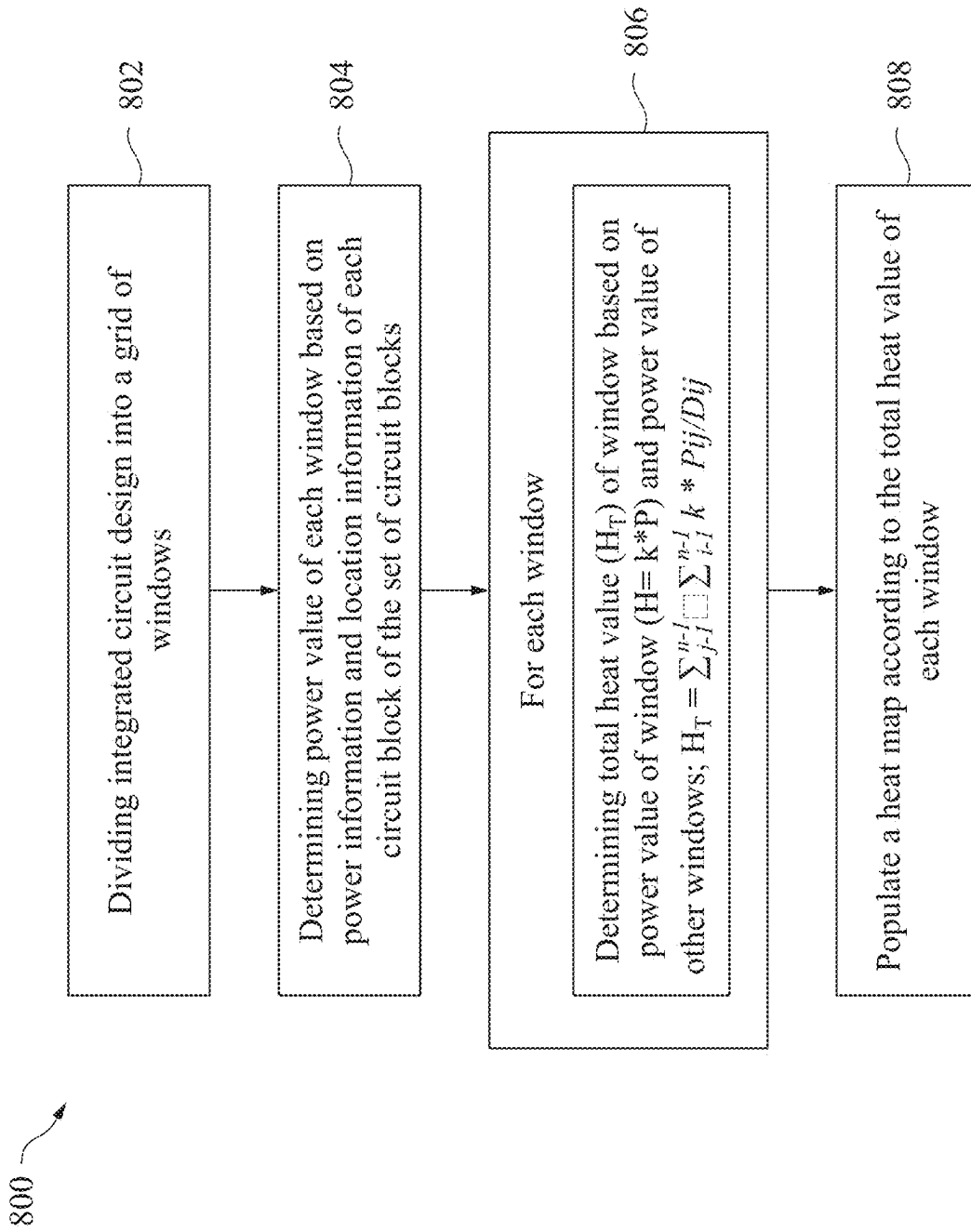


FIG. 8

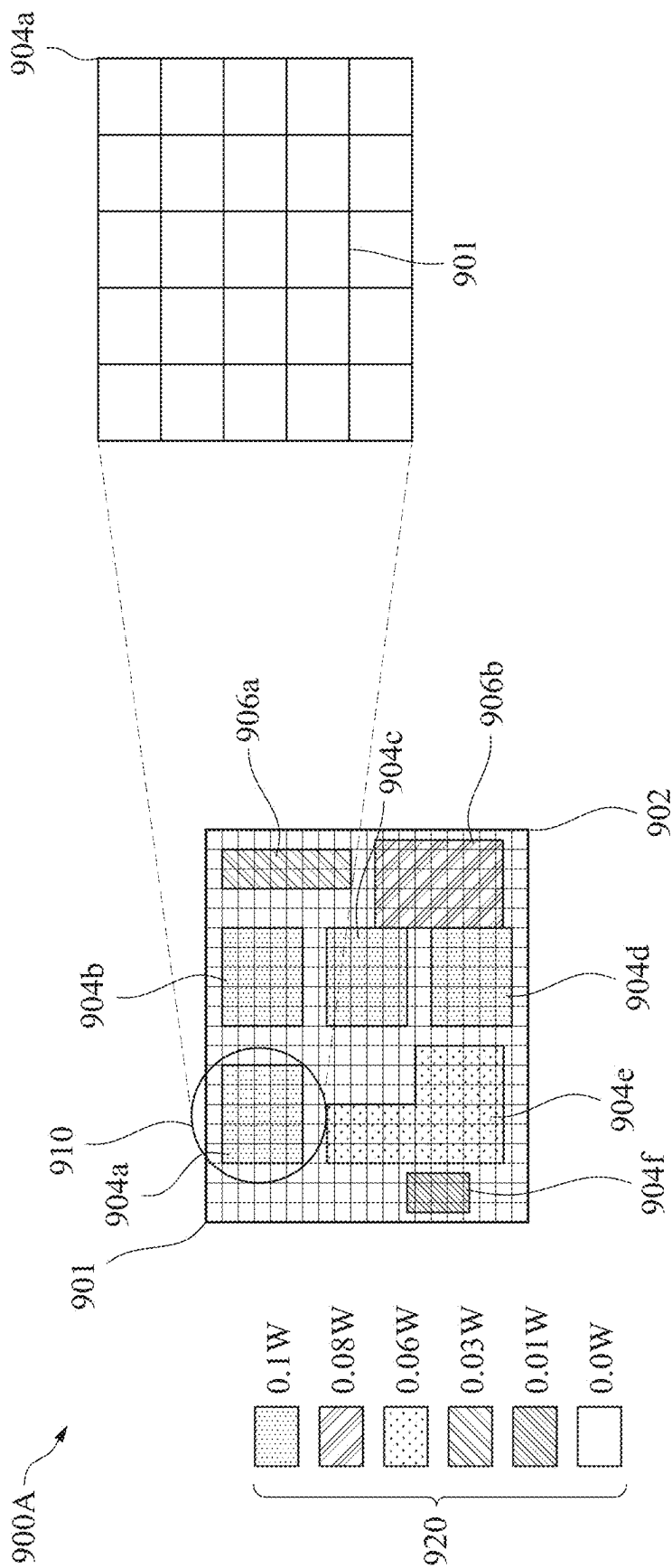


FIG. 9A

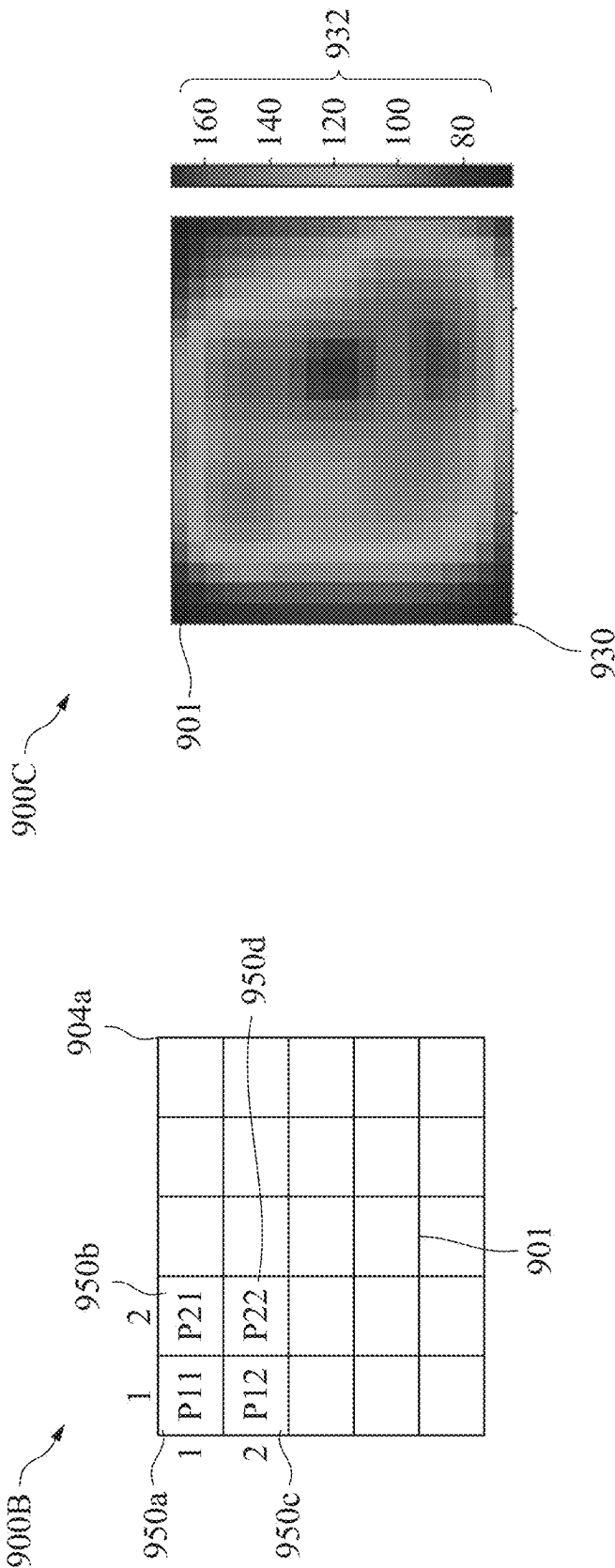


FIG. 9C

FIG. 9B

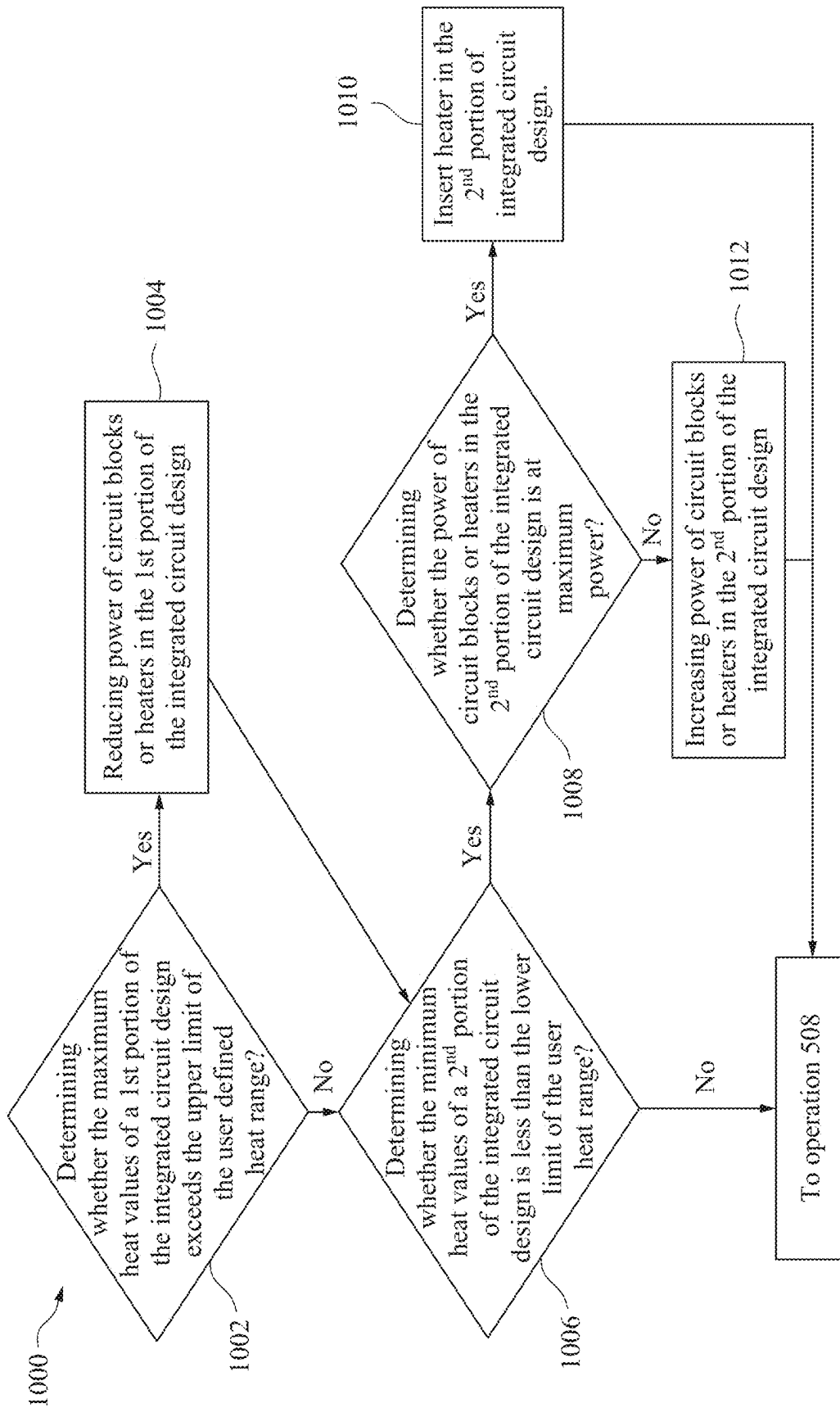
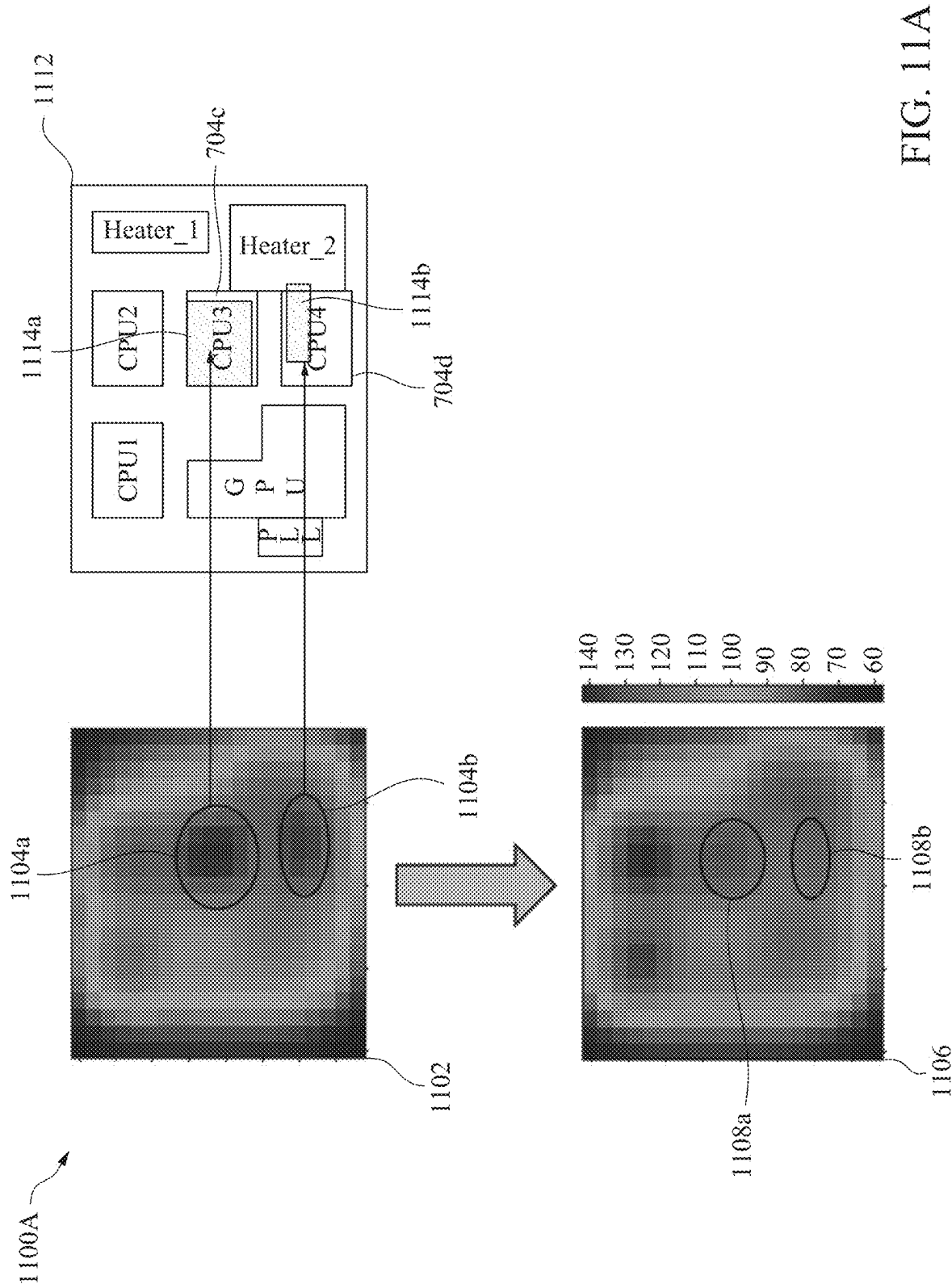
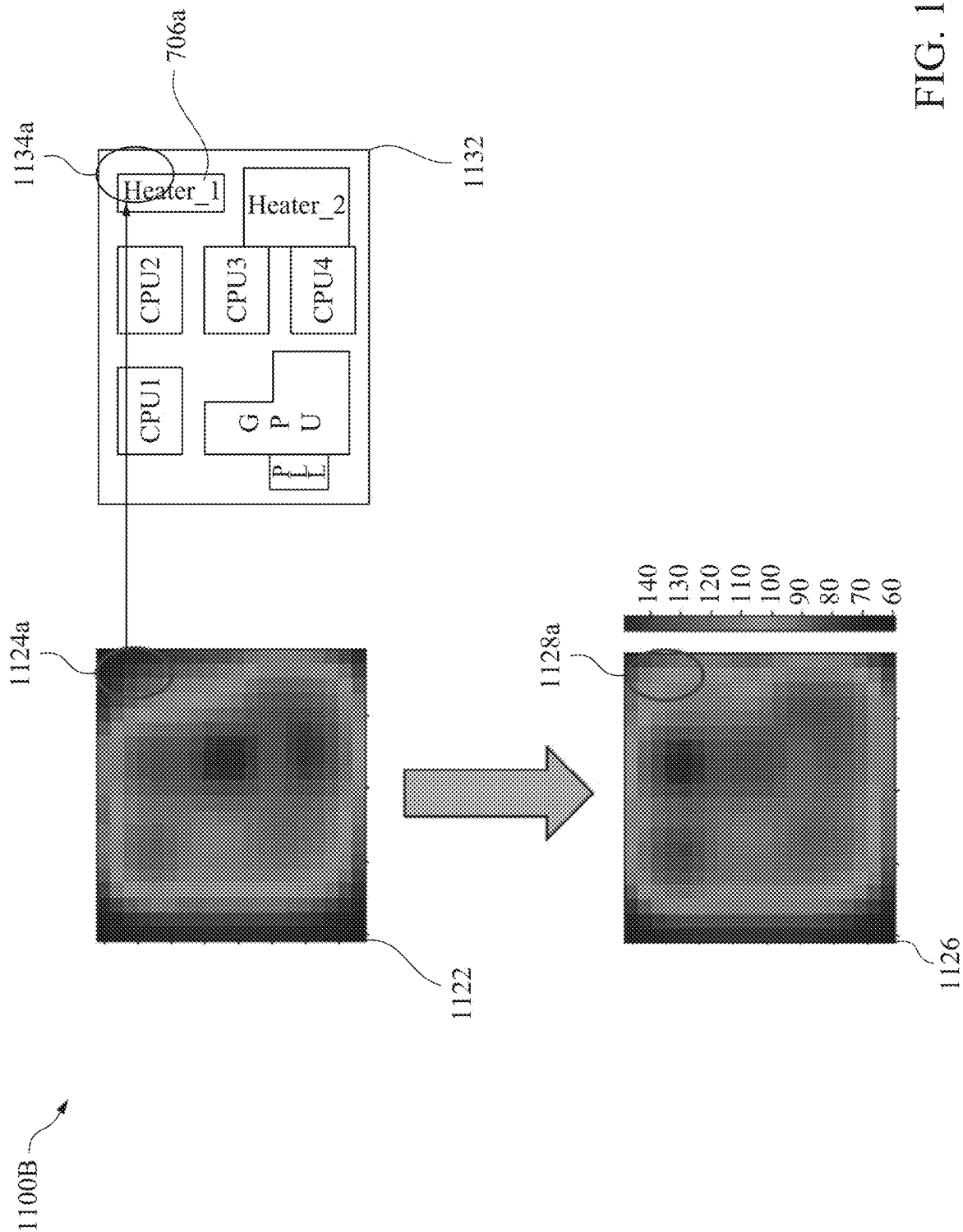
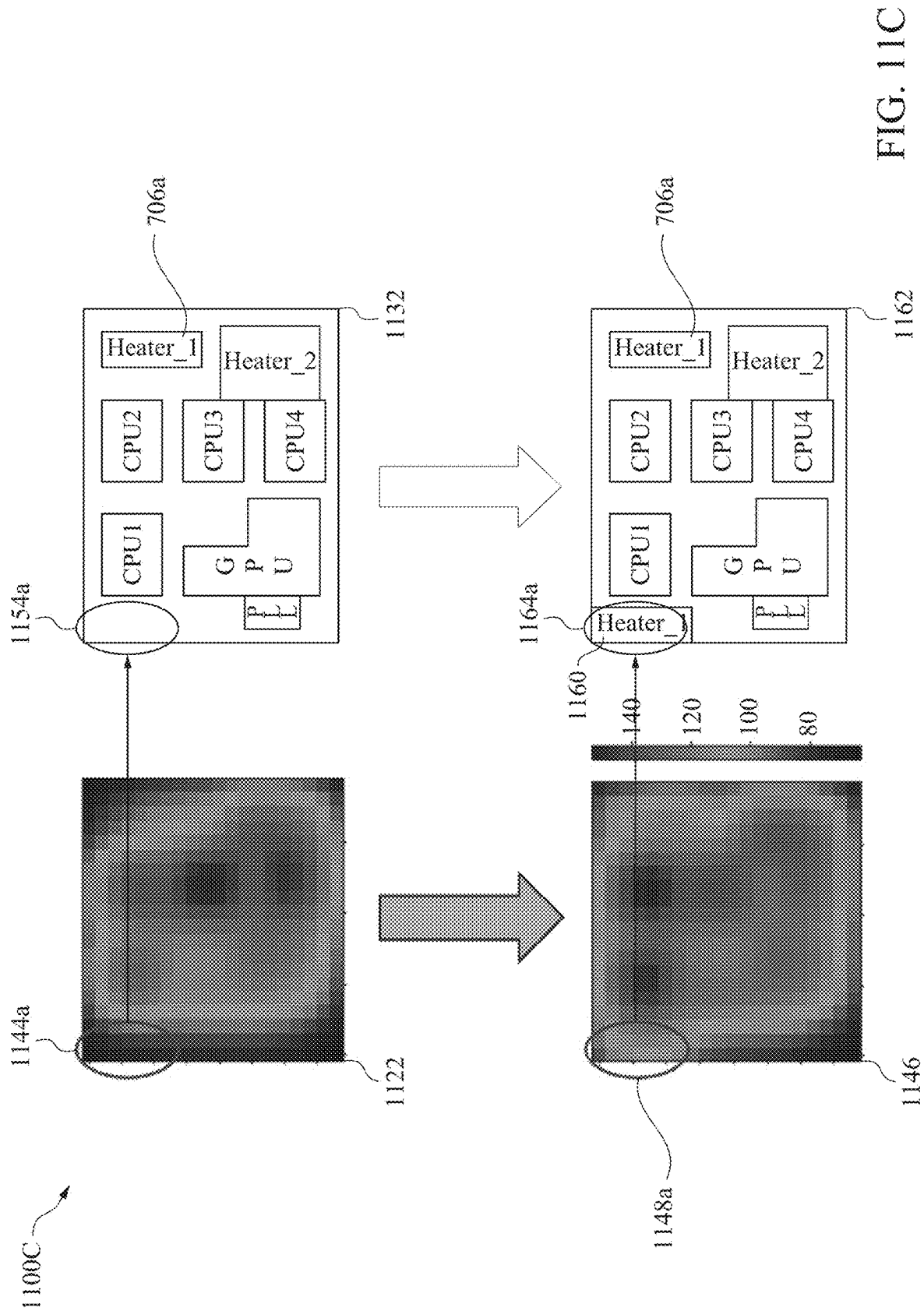


FIG. 10







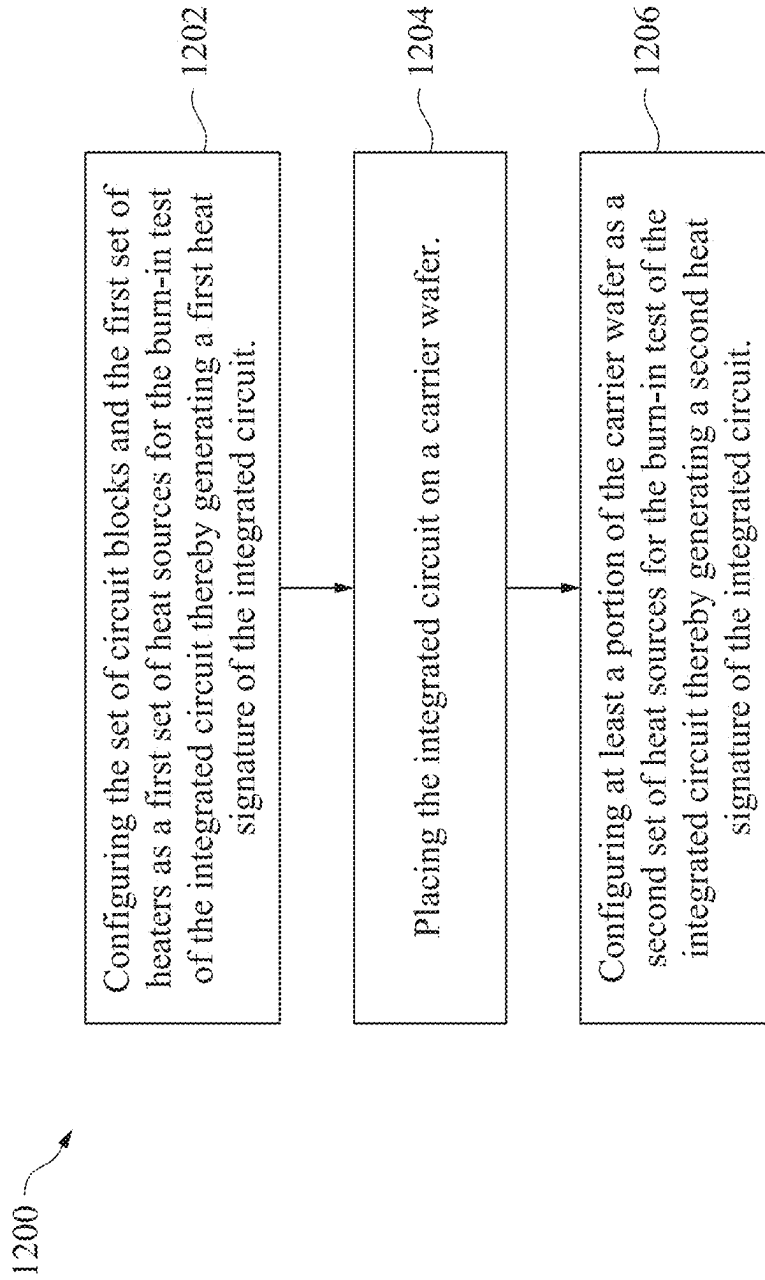


FIG. 12

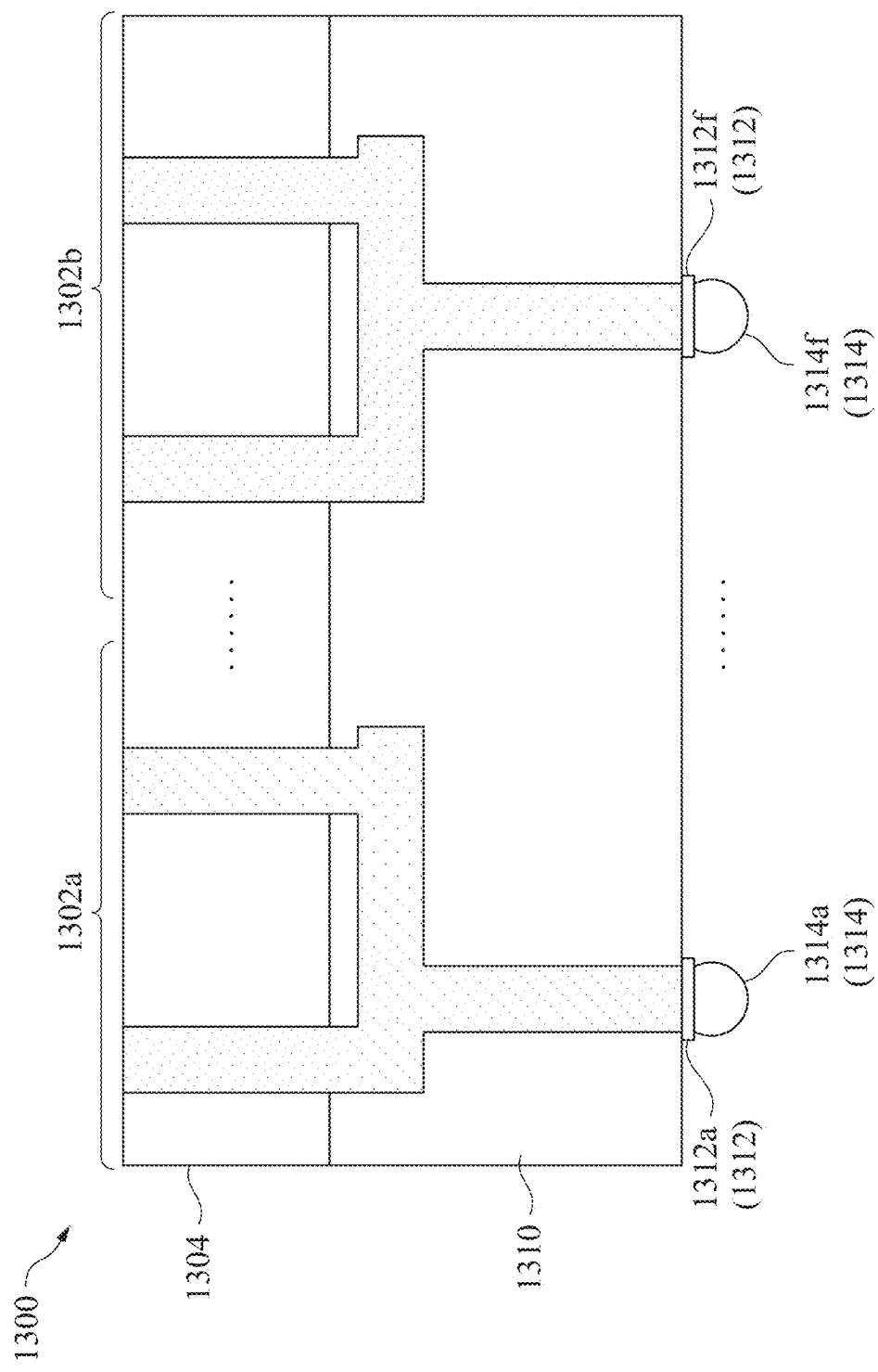


FIG. 13

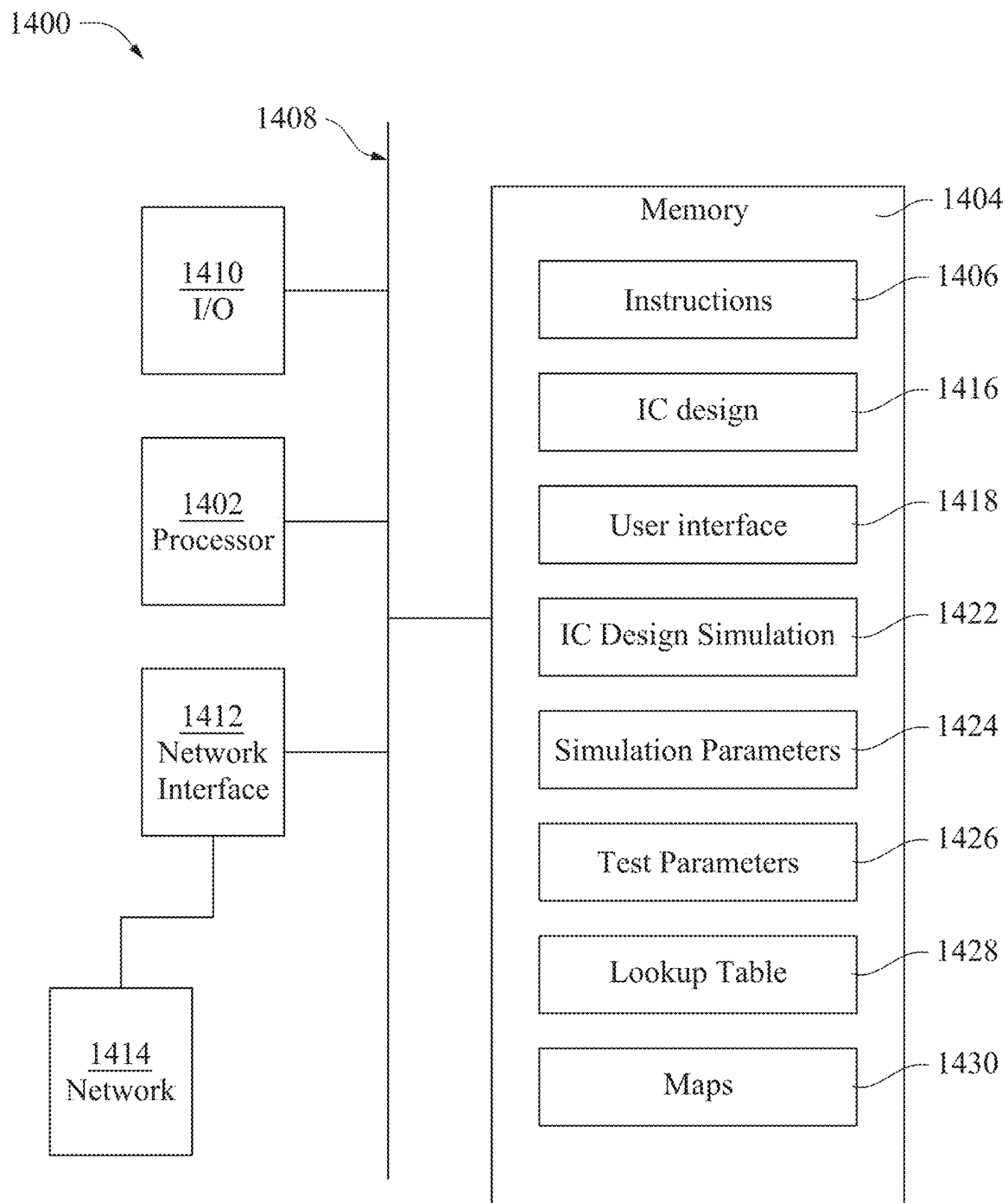


FIG. 14

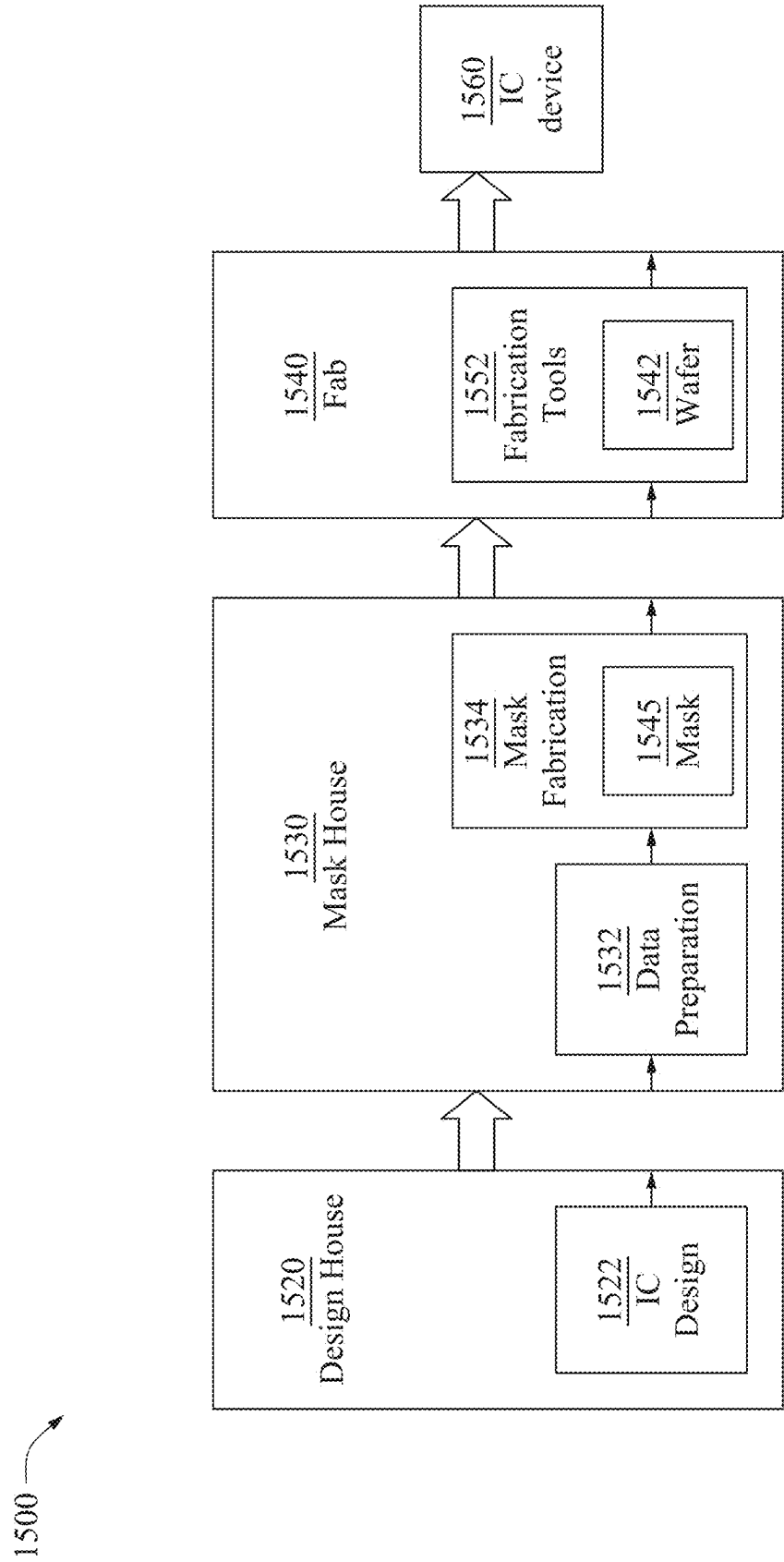


FIG. 15

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METHOD OF TESTING AN INTEGRATED CIRCUIT AND TESTING SYSTEM

PRIORITY CLAIM AND CROSS-REFERENCE

The present application is a continuation of U.S. application Ser. No. 17/393,232, filed Aug. 3, 2021, now U.S. Pat. No. 11,879,933, issued Jan. 23, 2024, which claims the priority of China Application No. 202110752449.3, filed Jul. 2, 2021, which are herein incorporated by reference in their entireties.

BACKGROUND

The recent trend in miniaturizing integrated circuits (ICs) has resulted in smaller devices which consume less power, yet provide functionality at higher speeds. The miniaturization process has also resulted in stricter design and manufacturing specifications. Various electronic design automation (EDA) tools generate, optimize and verify designs for semiconductor devices while ensuring that the design and manufacturing specifications are met. However, testing of semiconductor devices is a time consuming process.

BRIEF DESCRIPTION OF THE DRAWINGS

Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

FIG. 1 is a block diagram of a system, in accordance with some embodiments.

FIG. 2 is a diagram of a wafer, in accordance with some embodiments.

FIG. 3 is a diagram of a carrier wafer, in accordance with some embodiments.

FIG. 4 is a flowchart of a method of testing an integrated circuit, in accordance with some embodiments.

FIG. 5 is a flowchart of a method, in accordance with some embodiments.

FIG. 6 is a table, in accordance with some embodiments.

FIG. 7 is a block diagram of an integrated circuit design, in accordance with some embodiments.

FIG. 8 is a flowchart of a method of determining a heat signature of an integrated circuit design, in accordance with some embodiments.

FIG. 9A is a diagram of a power map, in accordance with some embodiments.

FIG. 9B is a block diagram of an integrated circuit design, in accordance with some embodiments.

FIG. 9C is a diagram of a heat map, in accordance with some embodiments.

FIG. 10 is a flowchart of a method of modifying an integrated circuit design, in accordance with some embodiments.

FIG. 11A is a diagram of power maps and an integrated circuit design, in accordance with some embodiments.

FIG. 11B is a diagram of power maps and an integrated circuit design, in accordance with some embodiments.

FIG. 11C is a diagram of power maps and an integrated circuit design, in accordance with some embodiments.

FIG. 12 is a flowchart of a method of simultaneously performing a burn-in test of the integrated circuit and an automated test of the integrated circuit, in accordance with some embodiments.

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FIG. 13 is a cross-sectional view of a heater, in accordance with some embodiments.

FIG. 14 is a schematic view of a system for designing an IC layout design, simulating an IC design, and manufacturing an IC circuit in accordance with some embodiments.

FIG. 15 is a block diagram of an IC manufacturing system, and an IC manufacturing flow associated therewith, in accordance with at least one embodiment of the present disclosure.

DETAILED DESCRIPTION

The following disclosure provides different embodiments, or examples, for implementing features of the provided subject matter. Specific examples of components, materials, values, steps, arrangements, or the like, are described below to simplify the present disclosure. These are, of course, merely examples and are not limiting. Other components, materials, values, steps, arrangements, or the like, are contemplated. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

In accordance with some embodiments, a method of testing an integrated circuit includes performing a simulation of a first heat distribution throughout an integrated circuit design, manufacturing the integrated circuit according to the integrated circuit design, and simultaneously performing a burn-in test of the integrated circuit and an automated test of the integrated circuit.

In some embodiments, by simultaneously performing a burn-in test of the integrated circuit and the automated test of the integrated circuit, the method of testing the integrated circuit has a shorter testing time than other approaches where burn-in testing is performed after the automated test. In some embodiments, after failure of the integrated circuit is detected, burn-in testing is stopped, thereby decreasing the burn-in testing time.

FIG. 1 is a block diagram of a system 100, in accordance with some embodiments.

In some embodiments, system 100 is a testing system configured to test a wafer 102. In some embodiments, system 100 is configured to test an integrated circuit 201 (shown in FIG. 2).

System 100 includes a wafer 102 coupled to each of test circuit board 104 and a carrier wafer 106. Wafer 102 includes a plurality of integrated circuits 201 (shown in FIG. 2).

Test circuit board **104** is configured to perform one or more tests of wafer **102**. Test circuit board **104** is electrically coupled to wafer **102**. In some embodiments, the test circuit board **104** is an automatic test equipment (ATE) board, and is configured to perform one or more automated tests of wafer **102**. Test circuit board **104** is electrically coupled between wafer **102** and a system **110**. Test circuit board **104** is electrically coupled to system **110** by a link **120**. In some embodiments, wafer **102** is electrically coupled to system **110** by test circuit board **104**. Other configurations of wafer **102** or test circuit board **104** are within the scope of the present disclosure.

Carrier wafer **106** is configured to carry the wafer **102**. Carrier wafer **106** includes one or more heater chips **108**. Carrier wafer **106** and heater chip **108** are electrically coupled to system **110** by a link **122**. In some embodiments, at least link **120** or **122** is a conductive wire. In some embodiments, at least link **120** or **122** is a data link configured to exchange data. In some embodiments, heater chip **108** includes one or more dies electrically coupled to system **110** by link **122**, and are configured to generate heat during a burn-in test of wafer **102**. Other configurations of carrier wafer **106** or heater chip **108** are within the scope of the present disclosure.

System **110** is electrically coupled to the integrated circuit within wafer **102** by test circuit board **104**. System **110** is electrically coupled to the one or more heater chips **108** of carrier wafer **106** by link **122**.

System **110** is configured to simultaneously perform a burn-in test of one or more integrated circuits in wafer **102** and one or more automated tests of one or more integrated circuits in wafer **102**. In some embodiments, the one or more automated tests of the one or more integrated circuits in wafer **102** are performed by test circuit board **104** and system **110**.

In some embodiments, a burn-in test includes testing wafer **102** for early failures of integrated circuits within wafer **102**. In some embodiments, the burn-in test is configured to apply thermal stress and environmental stress to the integrated circuits within wafer **102**, thus causing detectable failures in the integrated circuits within wafer **102**. In some embodiments, the failures are caused by faults in the manufacturing processes of wafer **102**. In some embodiments, the burn-in test is configured to apply a minimum burn-in temperature TBI over a duration of time. In some embodiments, the burn-in test is used to generate a burn-in heat distribution across the integrated circuits within wafer **102**. In some embodiments, the minimum burn-in temperature TBI ranges from about 120° Celsius (C) to about 160° C. In some embodiments, the duration of time of the burn-in test ranges from about 12 hours to about 72 hours.

In some embodiments, the one or more automated tests performed by test circuit board **104** and system **110** include voltage measurements, current measurements, timing measurements, reliability tests, or the like. In some embodiments, the one or more automated tests include operational tests of the one or more integrated circuits in wafer **102**.

In some embodiments, system **110** is an electronic design automation (EDA) tool configured to design and simulate performance of an integrated circuit design **700** (shown in FIG. 7) usable to manufacture integrated circuit **201** (FIG. 2).

Other configurations of system **110** are within the scope of the present disclosure.

In some embodiments, by simultaneously performing a burn-in test of one or more integrated circuits in wafer **102** and one or more automated tests (e.g., ATE) of one or more

integrated circuits in wafer **102**, system **100** has a shorter testing time than other approaches where burn-in testing is performed after ATE testing. For example, in some embodiments, after failure of integrated circuits in wafer **102** are detected, burn-in testing can be stopped, thereby decreasing the burn-in testing time.

In some embodiments, by configuring carrier wafer **106** with one or more heater chips **108**, heater chips **108** are configured as a heat source for burn-in testing of wafer **102**, and thus provides a uniform baking solution such that system **100** is capable of performing burn-in testing without a burn-in board or an oven thereby lowering cost compared to other approaches that use a burn-in board or an oven.

Other configurations of system **100** are within the scope of the present disclosure.

FIG. 2 is a diagram of a wafer **200**, in accordance with some embodiments.

Wafer **200** is an embodiment of wafer **102** of FIG. 1, and similar detailed description is therefore omitted. Components that are the same or similar to those in one or more of FIGS. 1-15 are given the same reference numbers, and detailed description thereof is thus omitted.

Wafer **200** includes a plurality of integrated circuits **201** arranged in an array.

A region **202** of wafer **200** corresponds to an integrated circuit **203** of the plurality of integrated circuits **201**. Integrated circuit **203** corresponds to a single integrated circuit of the plurality of integrated circuits **201**. In some embodiments, each integrated circuit of the plurality of integrated circuits **201** is the same. In some embodiments, at least one integrated circuit of the plurality of integrated circuits **201** is different from another integrated circuit of the plurality of integrated circuits **201**.

In some embodiments, the details of integrated circuit **203** are applicable to one or more of the plurality of integrated circuits **201**, and similar detailed description is omitted for brevity.

Other configurations of plurality of integrated circuits **201** are within the scope of the present disclosure.

Integrated circuit **203** includes a set of circuit blocks **204** and a set of heaters **206**.

Set of circuit blocks **204** includes at least a central processing unit (CPU) **204a**, a CPU **204b**, a CPU **204c**, a CPU **204d**, a graphics processing unit (GPU) **204e** or a phase locked loop (PLL) **204f**. Other numbers of circuits in set of circuit blocks **204** are within the scope of the present disclosure.

CPU, GPU and PLL are used for illustration, and other types of circuits in set of circuit blocks **204** are within the scope of various embodiments. For example, in some embodiments, set of circuit blocks **204** includes embedded processors including processor cores, digital signal processing (DSP) cores, embedded GPUs, interfaces such as universal serial bus (USB) controllers, ETHERNET, PCI-E, WIFI, WIMAX, or BLUETOOTH, peripherals such as universal asynchronous receiver transmitter (UART) or power management blocks, or memory modules and/or controllers.

At least CPU **204a**, CPU **204b**, CPU **204c**, CPU **204d**, GPU **204e** or PLL **204f** of the set of circuit blocks **204** is configured to generate heat by being operated. In some embodiments, at least CPU **204a**, CPU **204b**, CPU **204c**, CPU **204d**, GPU **204e** or PLL **204f** of the set of circuit blocks **204** is configured to generate heat by being operated during burn-in testing and ATE testing of integrated circuit **203**.

Set of heaters **206** includes at least a heater **206a** or a heater **206b**. At least heater **206a** or **206b** of the set of heaters **206** is configured to generate heat. In some embodiments, at least heater **206a** or **206b** of the set of heaters **206** is configured to generate heat during burn-in testing and ATE testing of integrated circuit **203**. In some embodiments, when the set of heaters **206** is not being used for testing, at least heater **206a** or **206b** of the set of heaters **206** is not configured to be operational, and thus does not generate heat.

In some embodiments, at least heater **206a** or **206b** of the set of heaters **206** includes one or more interconnects **1302a** and **1302b** (shown in FIG. 13). In some embodiments, at least heater **206a** or **206b** of the set of heaters **206** includes one or more circuit components configured to generate heat by being operated.

In some embodiments, the set of circuit blocks **204** and the set of heaters **206** are configured as a set of heat sources for at least the burn-in test or ATE tests performed by system **100**, and thereby generate a heat signature of integrated circuit **203**. In some embodiments, a heat signature corresponds to a map of a heat distribution throughout the integrated circuit. In some embodiments, an example of a heat distribution map **900C** is shown in FIG. 9C.

In some embodiments, the set of circuit blocks **204** and the set of heaters **206** are configured to generate a uniform heat distribution throughout integrated circuit **203**. In some embodiments, a uniform heat distribution corresponds to the heat distribution throughout the integrated circuit (e.g., integrated circuit **203**) that is within a heat range HR (shown in FIG. 5). In some embodiments, the heat range is defined or specified by a user of system **100** or method **500** (shown in FIG. 5). In some embodiments, the heat range is between a minimum heat value and a maximum heat value.

In some embodiments, the heat distribution throughout integrated circuit **203** or the set of integrated circuits **201** is modified by changing at least positions, number of circuit elements, a size or configured powers of the set of circuit blocks **204** and the set of heaters **206**. In some embodiments, at least the positions, number of circuit elements, size or configured powers of the set of circuit blocks **204** and the set of heaters **206** can be adjusted to generate a uniform heat distribution throughout integrated circuit **203**. In some embodiments, if integrated circuit **203** or the set of integrated circuits **201** have a uniform heat distribution, then system **100** is configured to perform burn-in testing while reducing a number of active elements in heater chip **108** of FIG. 1.

In some embodiments, the set of circuit blocks **204** and the set of heaters **206** are modified to generate a uniform heat distribution throughout integrated circuit **203** by operating at configured power levels that correspond to simulated design power levels (e.g., table **600** in FIG. 6). In some embodiments, in response to integrated circuit **203** or the set of integrated circuits **201** having a uniform heat distribution, system **100** is configured to perform burn-in testing without using active elements in heater chip **108** of FIG. 1.

Other configurations of the set of circuit blocks **204** and the set of heaters **206** are within the scope of the present disclosure.

Other configurations of wafer **200** are within the scope of the present disclosure.

FIG. 3 is a diagram of a carrier wafer **300**, in accordance with some embodiments.

Carrier wafer **300** is an embodiment of carrier wafer **106** of FIG. 1, and similar detailed description is therefore omitted.

Carrier wafer **300** includes a plurality of integrated circuit dies **301** arranged in an array.

A region **302** of carrier wafer **300** corresponds to an integrated circuit die **303** of the plurality of integrated circuit dies **301**. Integrated circuit die **303** corresponds to a single integrated circuit die of the plurality of integrated circuit dies **301**. In some embodiments, each integrated circuit die of the plurality of integrated circuit dies **301** is the same as another integrated circuit die of the plurality of integrated circuit dies. In some embodiments, at least one integrated circuit die of the plurality of integrated circuit dies **301** is different from another integrated circuit die of the plurality of integrated circuit dies **301**.

Region **302** has a same area as region **202** of wafer **200**. In some embodiments, region **302** has a different area as region **202** of wafer **200**.

Each integrated circuit die of the plurality of integrated circuit dies **301** is associated with each corresponding integrated circuit of the plurality of integrated circuits **201** of wafer **200**. In some embodiments, a position of each corresponding integrated circuit die of the plurality of integrated circuit dies **301** is in a same corresponding position as each corresponding integrated circuit of the plurality of integrated circuits **201** of wafer **200**.

Each integrated circuit die of the plurality of integrated circuit dies **301** has a same area as each integrated circuit of the plurality of integrated circuits **201** of wafer **200**. In some embodiments, at least one integrated circuit die of the plurality of integrated circuit dies **301** has a different area as at least one integrated circuit of the plurality of integrated circuits **201** of wafer **200**.

In some embodiments, the details of integrated circuit die **303** are applicable to one or more of the plurality of integrated circuit dies **301**, and similar detailed description is omitted for brevity.

Other configurations of the plurality of integrated circuit dies **301** are within the scope of the present disclosure.

Integrated circuit die **303** includes a set of circuit dies **304**. Set of circuit dies **304** is an array of dies that includes at least die **304a**, **304b**, . . . , **304y** or **304z**, where **z** is a positive integer corresponding to the number of dies in the set of circuit dies **304**.

Each die of the set of circuit dies **304** is the same as each other die of the set of circuit dies. In some embodiments, at least one die of the set of circuit dies **304** is different from at least another die of the set of circuit dies **304**.

Each die of the set of circuit dies **304** has a same area as each other die of the set of circuit dies. In some embodiments, at least one die of the set of circuit dies **304** has a different area as at least another die of the set of circuit dies **304**.

Each die of the set of circuit dies **304** is configured to operate as a corresponding heater of a set of heaters **306**.

Set of heaters **306** includes at least heater **306a**, **306b**, . . . , **306y** or **306z**. Each heater of the set of heaters **306** is configured to generate heat. In some embodiments, at least one or more heaters of the set of heaters **306** is configured to generate heat during burn-in testing and ATE testing of integrated circuit **203**.

In some embodiments, heat distribution throughout integrated circuit die **303** or the set of integrated circuit dies **301** is modified by adjusting configured powers or turning on or off the corresponding dies in the set of circuit dies **304**. In some embodiments, a sub-set of heaters in the set of heaters **306** is turned off by turning off a corresponding sub-set of dies of the set of dies **304**, and thus do not generate heat. In some embodiments, a sub-set of heaters in the set of heaters

306 is turned on by turning on a corresponding sub-set of dies of the set of dies **304**, and thus generate heat.

In some embodiments, the set of circuit dies **304** and the set of heaters **306** are configured as another set of heat sources for at least the burn-in test or ATE tests performed by system **100**, and thereby generate another heat signature of integrated circuit die **203**.

In some embodiments, the heat signature generated by integrated circuit **203** is combined with another heat signature of integrated circuit die **303**, to thereby generate a uniform heat distribution throughout integrated circuit **203**. In some embodiments, by generating a uniform heat distribution throughout integrated circuit **203**, burn-in testing performed on integrated circuit **203** and integrated circuit die **303** achieves one or more of the benefits discussed above in FIG. 1.

In some embodiments, integrated circuit die **303** or the set of integrated circuit dies **301** is configured as a supplemental heat source that when combined with the heat source of integrated circuit **203** or the set of integrated circuits **301** thereby generate a uniform heat distribution throughout integrated circuit **203**.

In some embodiments, by adjusting the number of operational or turned-on dies in the set of dies **304** adjusts the number of heaters in the set of heaters **306** thereby adjusting the amount of heat generated by integrated circuit die **303**, and thus integrated circuit die **303** functions as a supplemental heat source to the heat source of integrated circuit **203**.

Other configurations of the set of circuit dies **304** and the set of heaters **306** are within the scope of the present disclosure.

Other configurations of carrier wafer **300** are within the scope of the present disclosure.

FIG. 4 is a flowchart of a method **400** of testing an integrated circuit, in accordance with some embodiments.

In some embodiments, FIG. 4 is a flowchart of a method of operating system **100** of FIG. 1, system **1400** of FIG. 14 or IC manufacturing system **1500**.

It is understood that additional operations may be performed before, during, and/or after at least the method **400** depicted in FIG. 4, method **500** depicted in FIG. 5, method **800** depicted in FIG. 8, method **1000** depicted in FIG. 10, or method **1200** depicted in FIG. 12, and that some other operations may only be briefly described herein. In some embodiments, other order of operations of at least method **400**, method **500**, method **800**, method **1000**, or method **1200** is within the scope of the present disclosure. In some embodiments, one or more operations of at least method **400**, method **500**, method **800**, method **1000**, or method **1200** are not performed.

At least method **400**, method **500**, method **800**, method **1000**, or method **1200** includes exemplary operations, but the operations of at least method **400**, method **500**, method **800**, method **1000**, or method **1200** are not necessarily performed in the order shown. Operations of at least method **400**, method **500**, method **800**, method **1000**, or method **1200** may be added, replaced, changed order, and/or eliminated as appropriate, in accordance with the spirit and scope of disclosed embodiments. It is understood that at least method **400**, method **500**, method **800**, method **1000**, or method **1200** utilizes features of one or more of system **100**, wafer **200**, carrier wafer **300**, system **1400** or IC manufacturing system **1500**.

In operation **402** of method **400**, a simulation of a heat distribution throughout an integrated circuit design is performed by a processor. In some embodiments, the simulation

of operation **402** is a computer simulation performed by system **1400** of FIG. 14. In some embodiments, the processor of method **400** includes processor **1402** of FIG. 14.

In some embodiments, the integrated circuit design of method **400** includes at least integrated circuit design **700** of FIG. 7, integrated circuit design **900A** of FIG. 9A, integrated circuit design **1112** of FIG. 11A, integrated circuit design **1132** of FIGS. 11B-11C, or integrated circuit design **1162** of FIG. 11C. In some embodiments, the integrated circuit design of method **400** is a design of an integrated circuit.

In some embodiments, the heat distribution of method **400** includes at least the heat distribution shown by heat signature **930** of FIG. 9C, the heat distribution shown by heat signature **1102** of FIG. 11A, the heat distribution shown by heat signature **1106** of FIG. 11A, the heat distribution shown by heat signature **1122** of FIGS. 11B-11C, the heat distribution shown by heat signature **1126** of FIG. 11B or the heat distribution shown by heat signature **1146** of FIG. 11C.

In operation **404** of method **400**, an integrated circuit is manufactured according to the integrated circuit design. In some embodiments, the integrated circuit of method **400** includes at least wafer **102**, wafer **200**, plurality of integrated circuits **201** or integrated circuit **203**. In some embodiments, operation **404** is performed by IC manufacturing system **1500** of FIG. 15.

In operation **406** of method **400**, a burn-in test of the integrated circuit and an automated test of the integrated circuit are simultaneously performed. In some embodiments, operation **406** is performed by system **100** of FIG. 1. In some embodiments, the burn-in test of method **400** includes the burn-in test performed by system **100** of FIG. 1, and similar detailed description is omitted. In some embodiments, the automated test of method **400** includes the automated or ATE test performed by system **100** of FIG. 1, similar detailed description is omitted.

Method **400** operates to achieve the benefits discussed above with respect to at least system **100** of FIG. 1, wafer **200** of FIG. 2 or carrier wafer **300** of FIG. 3.

FIG. 5 is a flowchart of a method **500**, in accordance with some embodiments.

In some embodiments, method **500** includes operations of method **400** of FIG. 4, and similar detailed description is omitted. For example, in some embodiments, method **500** includes an embodiment of operation **402** and **404** of FIG. 4.

In some embodiments, method **500** is a method of performing a simulation of a heat distribution throughout an integrated circuit design, and a method of manufacturing an integrated circuit according to the integrated circuit design.

In some embodiments, the method of performing the simulation of the heat distribution throughout the integrated circuit design includes operations **502**, **504**, **506**, **508**, **510**, **514** and **516**. In some embodiments, operations **502**, **504**, **506**, **508**, **510**, **514** and **516** are an embodiment of operation **402** of FIG. 4, and similar detailed description is omitted. In some embodiments, one or more of operations **502**, **504**, **506**, **508**, **510**, **514** and **516** are performed by system **110** or system **1400** of FIG. 14.

In some embodiments, the method of manufacturing the integrated circuit according to the integrated circuit design includes operation **512**. In some embodiments, operation **512** is operation **404** of FIG. 4, and similar detailed description is omitted. In some embodiments, operation **512** is performed by IC manufacturing system **1500** of FIG. 15.

In operation **502** of method **500**, a heat range HR of an integrated circuit design is received from a user. In some embodiments, the user of method **500** includes the user of at

least system **100**, system **110** or system **1400** of FIG. **14**. In some embodiments, the heat range HR of method **500** includes the heat range of FIG. **1**, and similar detailed description is omitted. In some embodiments, the heat range HR is defined between a minimum heat value Hmin and a maximum heat value Hmax.

In some embodiments, heat values included within the heat range HR are considered as a uniform heat distribution throughout the integrated circuit design. In some embodiments, the integrated circuit design of method **500** includes at least integrated circuit design **700** of FIG. **7**, integrated circuit design **900A** of FIG. **9A**, integrated circuit design **1112** of FIG. **11A**, integrated circuit design **1132** of FIGS. **11B-11C**, or integrated circuit design **1162** of FIG. **11C**.

In some embodiments, the integrated circuit design of method **500** is a design of an integrated circuit, such as at least wafer **102**, wafer **200**, plurality of integrated circuits **201** or integrated circuit **203**.

In operation **504** of method **500**, configured power information for a set of circuit blocks and a set of heaters in the integrated circuit design are obtained. In some embodiments, operation **504** of method **500** includes obtaining configured power information for each circuit block of the set of circuit blocks and each heater of the set of heaters in the integrated circuit design.

In some embodiments, the configured power information for the set of circuit blocks and the set of heaters in the integrated circuit design are obtained from a table, such as table **600** of FIG. **6**. In some embodiments, table **600** is stored in memory **1404** of FIG. **14**. In some embodiments, the configured power information of method **500** includes at least the set of maximum configured power entries **604** of FIG. **6** or set of configured power entries **606** of FIG. **6**.

In some embodiments, the set of circuit blocks of the integrated circuit design of method **500** includes at least the set of circuit blocks **704** of FIG. **7** or FIGS. **11A-11C**. In some embodiments, the set of circuit blocks of the integrated circuit design of method **500** includes at least circuit block **602a**, **602b** or **602e** of FIG. **6**.

In some embodiments, the set of heaters of the integrated circuit design of method **500** includes at least set of heaters **706** of FIG. **7** or heater **1160** of FIG. **11C**. In some embodiments, the set of heaters of the integrated circuit design of method **500** includes at least heater **602c** or **602d** of FIG. **6**.

In operation **506** of method **500**, location information for the set of circuit blocks and the set of heaters in the integrated circuit design are extracted from a design file (DEF). In some embodiments, operation **506** of method **500** includes extracting location information for each circuit block of the set of circuit blocks and each heater of the set of heaters in the integrated circuit design from the design file. In some embodiments, the design file of method **500** is stored in memory **1404** of FIG. **14**.

In some embodiments, the location information for the set of circuit blocks and the set of heaters in the integrated circuit design of method **500** include the positions of each element in the integrated circuit design. In some embodiments, the design file of method **500** includes a design exchange format (DEF) that represents a physical layout of the integrated circuit design. In some embodiments, the DEF is in an American Standard Code for Information Interchange (ASCII) format, and represents a netlist and circuit layout of the integrated circuit design.

In operation **508** of method **500**, a heat signature of the integrated circuit design is determined from the configured

power information and the location information for the set of circuit blocks and the set of heaters included in the integrated circuit design.

In some embodiments, the heat signature is a two dimensional map of heat values HV arranged throughout the integrated circuit design. In some embodiments, the heat values HV are generated by each circuit block of the set of circuit blocks and each heater of the set of heaters included in the integrated circuit design.

In some embodiments, the heat values HV of the integrated circuit design are determined from formula 2 (described below in FIGS. **8** & **9A-9C**). In some embodiments, method **800** of FIG. **8** is an embodiment of operation **508**.

In some embodiments, operation **508** of method **500** includes determining the heat signature of the integrated circuit design from the configured power information and the location information for each circuit block of the set of circuit blocks and each heater of the set of heaters included in the integrated circuit design.

In some embodiments, the heat signature of method **500** includes at least heat signature **930** of FIG. **9C**, heat signature **1102** of FIG. **11A**, heat signature **1106** of FIG. **11A**, heat signature **1122** of FIGS. **11B-11C**, heat signature **1126** of FIG. **11B** or heat signature **1146** of FIG. **11C**.

In operation **510** of method **500**, a determination is made whether heat values HV of the heat signature of the integrated circuit design are within the heat range HR specified by the user.

In some embodiments, operation **510** includes determining whether heat values HV of the heat signature of the integrated circuit design are equal to or greater than the minimum heat value Hmin and equal to or less than the maximum heat value Hmax.

In some embodiments, if the heat values HV of the heat signature of the integrated circuit design are within the heat range HR specified by the user, then the result of operation **510** is a “yes”, and method **500** proceeds to operation **512**. In some embodiments, if the heat values HV of the heat signature of the integrated circuit design are within the heat range HR specified by the user, then the heat values HV of the heat signature of the integrated circuit design are sufficient to cause the integrated circuit design to generate a uniform heat distribution, thus indicating that an integrated circuit manufactured based on the integrated circuit design also generates a uniform heat distribution, and the burn-in test of operation **406** performed by system **100** is performed without enabling the heaters of carrier wafer **106**.

In some embodiments, if the heat values HV of the heat signature of the integrated circuit design are not within the heat range HR specified by the user, then the result of operation **510** is a “no”, and method **500** proceeds to operation **514**.

In operation **512** of method **500**, an integrated circuit is manufactured according to the integrated circuit design. In some embodiments, the integrated circuit of method **500** includes at least wafer **102**, wafer **200**, plurality of integrated circuits **201** or integrated circuit **203**. In some embodiments, operation **512** is performed by IC manufacturing system **1500** of FIG. **15**. In some embodiments, operation **512** is operation **404** of FIG. **4**, and similar detailed description is omitted.

In operation **514** of method **500**, a determination is made whether a number of iterations of operation **508**, **510** or **516** exceeds a user defined limit. In some embodiments, the user defined limit of method **500** includes a maximum number of iterations that at least operation **508**, **510** or **516** is per-

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formed. In operation **502** of method **500**, the user defined limit of method **500** is received from the user.

In some embodiments, if the number of iterations of operation **508**, **510** or **516** does not exceed the user defined limit, then the result of operation **514** is a “no”, and method **500** proceeds to operation **516**. In some embodiments, if the number of iterations of operation **508**, **510** or **516** does not exceed the user defined limit, then the heat values HV of the heat signature of the integrated circuit design are insufficient to cause the integrated circuit design to generate a uniform heat distribution, and method **500** attempts to modify the integrated circuit design by operation **516**.

In some embodiments, if the number of iterations of operation **508**, **510** or **516** exceeds the user defined limit, then the result of operation **514** is a “yes”, and method **500** proceeds to operation **512**. In some embodiments, if the number of iterations of operation **508**, **510** or **516** exceeds the user defined limit, then the heat values HV of the heat signature of the integrated circuit design are insufficient to cause the integrated circuit design to generate a uniform heat distribution, thus indicating that an integrated circuit manufactured based on the integrated circuit design also does not generate a uniform heat distribution, and therefore when the burn-in test of operation **406** performed by system **100** is performed, the heaters of carrier wafer **106** are enabled thereby generating a supplemental heat distribution that when combined with the heat distribution of the integrated circuit is a uniform heat distribution.

In operation **516** of method **500**, the integrated circuit design is modified. In some embodiments, the integrated circuit design is modified in response to determining that the heat values HV of the heat signature of the integrated circuit design are not within the heat range HR. In some embodiments, method **1000** of FIG. **10** is an embodiment of operation **516**.

In some embodiments, operation **516** includes one or more of adding a new circuit block to the set of circuit blocks, adding a new heater to the set of heaters, removing a first heater from the set of heaters, moving a position of the first heater of the set of heaters, modifying the configured power of a first circuit block of the set of circuit blocks, modifying the configured power of the first heater of the set of heaters or modifying a size of the first heater of the set of heaters.

In some embodiments, the new circuit block or the first circuit block of the set of circuit blocks of method **500** includes one or more circuit blocks similar to the set of circuit blocks **704** of the integrated circuit design. In some embodiments, the new heater or the first heater of the set of heaters of method **500** includes one or more heaters similar to the set of heaters **706** or heater **1160** of FIG. **11C** of the integrated circuit design.

In some embodiments, one or more of operations **508**, **510**, **514** or **516** are repeated thereby causing the integrated circuit design to be modified. In some embodiments, one or more of operations **508**, **510**, **514** or **516** are repeated until the heat values HV of the heat signature of the integrated circuit design are within the heat range HR specified by the user or the number of iterations of operation **508**, **510** or **516** exceeds the user defined limit.

Method **500** operates to achieve the benefits discussed above with respect to at least system **100** of FIG. **1**, wafer **200** of FIG. **2** or carrier wafer **300** of FIG. **3**.

FIG. **6** is a table **600**, in accordance with some embodiments.

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Table **600** is a lookup table of parameters of an integrated circuit design, such as integrated circuit design **700** of FIG. **7**.

Table **600** is utilized with one or more operations of method **500** of FIG. **5**. In some embodiments, method **500** is utilized with lookup table **600**. In some embodiments, one or more of operations **502**, **504**, **506**, **508** or **516** of method **500** utilize a lookup table similar to table **600** of FIG. **6**. In some embodiments, table **600** is stored in memory **1404** of FIG. **14**. In some embodiments, table **600** is generated by system **1400** of FIG. **14**.

Table **600** comprises 6 rows and 3 columns. Column 1 comprises a set of IC design elements **602**. Column 2 comprises a set of maximum configurable power entries **604** for the corresponding set of IC design elements **602**. Column 3 comprises a set of configurable power entries **606** for the corresponding set of IC design elements **602**. Each entry in column 1 has a corresponding entry in column 2 and a corresponding entry in column 3.

Other numbers of columns or rows in table **600** are within the scope of the present disclosure.

The set of IC design elements **602** includes at least IC design elements **602a**, **602b**, **602c**, **602d** or **602e**. The set of IC design elements **602** corresponds to the set of circuit blocks **704** or the set of heaters **706** of the integrated circuit design. IC design elements **602a**, **602b** and **602e** correspond to one or more circuit blocks in the set of circuit blocks **704** of integrated circuit design **700**. IC design elements **602c** and **602d** correspond to one or more heaters in the set of heaters **706** of integrated circuit design **700**.

Other numbers of IC design elements in the set of IC design elements **602** are within the scope of the present disclosure. Other numbers of circuit blocks or heaters in the set of IC design elements **602** are within the scope of the present disclosure.

Column 2 comprises a set of maximum configurable power entries **604** corresponding to the set of IC design elements **602**. The set of maximum configurable power entries **604** includes at least maximum configurable power entries **604a**, **604b**, **604c**, **604d** or **604e**. Each maximum configurable power entry **604a**, **604b**, **604c**, **604d** or **604e** of the set of maximum configurable power entries **604** is the maximum configurable power at which the corresponding IC design element **602a**, **602b**, **602c**, **602d** or **602e** of the set of IC design elements **602** can operate.

Other numbers or values of maximum configurable power entries in the set of maximum configurable power entries **604** are within the scope of the present disclosure.

Column 3 comprises a set of configurable power entries **606** corresponding to the set of IC design elements **602** or the set of maximum configurable power entries **604**.

The set of configurable power entries **606** includes at least configurable power entries **606a**, **606b**, **606c**, **606d** or **606e**. Each configurable power entry **606a**, **606b**, **606c**, **606d** or **606e** of the set of configurable power entries **606** is the configurable power that the corresponding IC design element **602a**, **602b**, **602c**, **602d** or **602e** of the set of IC design elements **602** can operate at (without including the maximum configurable powers of column 2).

Other numbers or values of configurable power entries **606** in the set of configurable power entries **606** are within the scope of the present disclosure.

Configurable power entry **606a** includes one or more configurable power entries at which IC design element **602a** can operate. For example, configurable power entry **606a** includes configurable power entries 6 Watts (W), 4 W and 2 W at which IC design element **602a** can operate.

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Configurable power entry **606b** includes one or more configurable power entries at which IC design element **602b** can operate. For example, configurable power entry **606b** includes configurable power entries 6 W, 4 W and 2 W at which IC design element **602b** can operate.

Configurable power entry **606c** includes one or more configurable power entries at which IC design element **602c** can operate. For example, configurable power entry **606c** includes configurable power entries 1 W and 0.5 W at which IC design element **602c** can operate.

Configurable power entry **606d** includes one or more configurable power entries at which IC design element **602d** can operate. For example, configurable power entry **606d** includes configurable power entries 2.5 W and 1.25 W at which IC design element **602d** can operate.

Configurable power entry **606d** includes one or more configurable power entries at which IC design element **602d** can operate. For example, configurable power entry **606d** includes configurable power entries 2.5 W and 1.25 W at which IC design element **602d** can operate.

Configurable power entry **606e** includes one or more configurable power entries at which IC design element **602e** can operate. For example, configurable power entry **606e** includes configurable power entry 0.25 W at which IC design element **602e** can operate.

In some embodiments, during operation **516** of method **500**, the integrated circuit design is modified by adjusting (e.g., increasing or decreasing) the configured power of one or more circuit blocks of the set of circuit blocks or one or more heaters of the set of heaters according to the set of maximum configurable power entries **604** and the set of configurable power entries **606** of table **600**. For example, in some embodiments, if the heat values HV generated by IC design element **602a** with a configurable power entry **606a** of 2 W is less than the heat range HR, then the configurable power entry **606a** of IC design element **602a** can be increased from 2 W to 4 W up to the maximum configurable power entry **604a** in column 2.

FIG. 7 is a block diagram of an integrated circuit design **700**, in accordance with some embodiments.

Integrated circuit design **700** corresponds to the integrated circuit design associated with table **600** of FIG. 6, and similar detailed description is therefore omitted. In some embodiments, integrated circuit design **700** is an integrated circuit design that corresponds to the set of IC design elements **602** of table **600** of FIG. 6, the set of maximum configurable power entries **604**, and the set of configurable power entries **606**, and similar detailed description is therefore omitted.

In some embodiments, integrated circuit design **700** is a design of integrated circuit **203** of FIG. 2 or one or more of the integrated circuits of the plurality of integrated circuits **201** of FIG. 2, and similar detailed description is therefore omitted. In some embodiments, integrated circuit design **700** is usable to manufacture integrated circuit **203** of FIG. 2, and similar detailed description is therefore omitted. In some embodiments, integrated circuit design **700** is stored in memory **1404** of FIG. 14. In some embodiments, integrated circuit design **700** is generated by system **1400** of FIG. 14.

Integrated circuit design **700** includes a set of circuit blocks **704** and a set of heaters **706**. In some embodiments, set of circuit blocks **704** of integrated circuit design **700** is usable to manufacture set of circuit blocks **204** of integrated circuit **203** of FIG. 2, and similar detailed description is therefore omitted. In some embodiments, set of heaters **706** of integrated circuit design **700** is usable to manufacture set of heaters **206** of integrated circuit **203** of FIG. 2, and similar

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detailed description is therefore omitted. In some embodiments, the operation of the elements of the integrated circuit design **700** are similar to the corresponding elements of the integrated circuit **203**, and similar description is omitted for brevity.

Set of circuit blocks **704** includes at least a CPU **704a**, a CPU **704b**, a CPU **704c**, a CPU **704d**, a GPU **704e** or a PLL **704f**. In some embodiments, at least CPU **704a**, CPU **704b**, CPU **704c**, CPU **704d**, GPU **704e** or PLL **704f** of integrated circuit design **700** is usable to manufacture at least CPU **204a**, CPU **204b**, CPU **204c**, CPU **204d**, GPU **204e** or PLL **204f** of integrated circuit **203** of FIG. 2, and similar detailed description is therefore omitted.

In some embodiments, at least CPU **704a**, CPU **704b**, CPU **704c**, CPU **704d**, GPU **704e** or PLL **704f** of the set of circuit blocks **704** is configured to operate at one or more power levels according to the set of configurable power entries **606** of table **600** or the set of maximum configurable power entries **604** of table **600** thereby generating heat values and the heat signature of the integrated circuit design during execution of operation **402** of method **400**.

In some embodiments, at least CPU **204a**, CPU **204b**, CPU **204c**, CPU **204d**, GPU **204e** or PLL **204f** of the set of circuit blocks **204** of FIG. 2 is configured to operate at one or more power levels according to the set of configurable power entries **606** of table **600** or the set of maximum configurable power entries **604** of table **600** thereby generating heat during burn-in testing and ATE testing (e.g., operation **406** of method **400**) of integrated circuit **203**.

Other numbers of circuits in set of circuit blocks **704** are within the scope of the present disclosure.

Set of heaters **706** includes at least a heater **706a** or a heater **706b**. In some embodiments, at least heater **706a** or heater **706b** of integrated circuit design **700** is usable to manufacture at least heater **206a** or heater **206b** of integrated circuit **203** of FIG. 2, and similar detailed description is therefore omitted.

In some embodiments, at least heater **706a** or heater **706b** of the set of heaters **706** is configured to operate at one or more power levels according to the set of configurable power entries **606** of table **600** or the set of maximum configurable power entries **604** of table **600** thereby generating heat values and the heat signature of the integrated circuit design during execution of operation **402** of method **400**.

In some embodiments, at least heater **206a** or heater **206b** of the set of heaters **206** of FIG. 2 is configured to operate at one or more power levels according to the set of configurable power entries **606** of table **600** or the set of maximum configurable power entries **604** of table **600** thereby generating heat during burn-in testing and ATE testing (e.g., operation **406** of method **400**) of integrated circuit **203**.

Other numbers of heaters in set of heaters **706** are within the scope of the present disclosure.

In some embodiments, the set of circuit blocks **704** and the set of heaters **706** are configured as a set of heat sources, and thereby generate a heat signature of integrated circuit design **700** during execution of operation **402** of method **400**.

In some embodiments, the set of circuit blocks **704** and the set of heaters **706** are configured to generate a uniform heat distribution throughout integrated circuit design **700**. In some embodiments, the set of circuit blocks **704** and the set of heaters **706** are modified to generate a uniform heat distribution throughout integrated circuit design **700** by operating at configured power levels contained in table **600** of FIG. 6.

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In some embodiments, the heat distribution throughout integrated circuit design **700** is modified by changing at least positions, number of circuit elements, a size or configured powers of the set of circuit blocks **704** and the set of heaters **706**. In some embodiments, at least the positions, number of circuit elements, size or configured powers of the set of circuit blocks **704** and the set of heaters **706** can be adjusted to generate a uniform heat distribution throughout integrated circuit design **700**.

In some embodiments, if integrated circuit design **700** has a uniform heat distribution, then an integrated circuit **201**, manufactured based on integrated circuit design **700**, is useable with system **100** while reducing the number of active (e.g., turned-on) elements in heater chip **108** of FIG. **1** during burn-in testing of integrated circuit **201** or wafer **102**. In some embodiments, the number of active elements in heater chip **108** of FIG. **1** during burn-in testing of wafer **102** or integrated circuit **201** based on integrated circuit design **700** is zero.

In some embodiments, if integrated circuit design **700** does not have a uniform heat distribution, then an integrated circuit **201**, manufactured based on integrated circuit design **700**, is combined with a supplemental heat source (e.g., generated by a number of turned-on or active elements in heater chip **108** of FIG. **1**) during burn-in testing of integrated circuit **201** or wafer **102** performed by system **100**.

Other configurations of the set of circuit blocks **704** and the set of heaters **706** are within the scope of the present disclosure.

Other configurations of integrated circuit design **700** are within the scope of the present disclosure.

FIG. **8** is a flowchart of a method **800** of determining a heat signature of an integrated circuit design, in accordance with some embodiments.

In some embodiments, method **800** is an embodiment of operation **508** of method **500**, and similar detailed description is omitted. In some embodiments, one or more of operations **802**, **804**, **806** and **808** are performed by system **110** or system **1400** of FIG. **14**.

In some embodiments, method **800** is useable to determine a heat signature of an integrated circuit design such as integrated circuit design **700**, integrated circuit design **902** of FIG. **9A**, integrated circuit design **1112** of FIG. **11A**, integrated circuit design **1132** of FIGS. **11B-11C**, or integrated circuit design **1162** of FIG. **11C**.

In some embodiments, method **800** is useable to determine a heat signature such as heat signature **930** of FIG. **9C**, heat signatures **1102** and **1106** of FIG. **11A**, heat signatures **1122** and **1126** of FIG. **11B**, and heat signatures **1122** and **1146** of FIG. **11C**.

In operation **802** of method **800**, an integrated circuit design is divided into an array of windows. In some embodiments, the integrated circuit design is divided into the array of windows based on a window size. In some embodiments, the window size is defined by a user. In some embodiments, the window size of window of the array of windows is the same. In some embodiments, the window size ranges from about 1 μm to about 20 μm . Other values of window size are within the scope of the present disclosure.

In some embodiments, the integrated circuit design of method **800** includes at least integrated circuit design **700** of FIG. **7**, integrated circuit design **902** of FIG. **9A**, integrated circuit design **1112** of FIG. **11A**, integrated circuit design **1132** of FIGS. **11B-11C**, or integrated circuit design **1162** of FIG. **11C**.

In some embodiments, the array of windows of method **800** includes at least the array of windows **901** of FIGS.

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9A-9B. In some embodiments, the integrated circuit design **902** of FIG. **9A** includes an array of windows **901** that illustrates a non-limiting example of an array of windows after execution of operation **802**.

In some embodiments, each window of the array of windows has a square shape. In some embodiments, one or more windows of the array of windows is circular, triangular, rectangular, square, hexagonal, or other geometric shapes. In some embodiments, one or more windows of the array of windows is a polygon. Other shapes of the array of windows are within the scope of the present disclosure.

In operation **804** of method **800**, the power value of each window in the array of windows is determined based on power information and location information of the set of circuit blocks and the set of heaters. In some embodiments, operation **804** includes generating an integrated circuit power map similar to integrated circuit power map **900A** in FIG. **9A**. In some embodiments, the power value of each window of method **800** corresponds to the power consumed by each circuit block or heater within the corresponding window.

In some embodiments, the power information of method **800** includes the configured power information obtained for operation **504** of method **500**. In some embodiments, the power information of method **800** includes the set of configured power entries **606** from table **600** or the set of maximum configured power entries **604** from table **600**. In some embodiments, the location information of method **800** includes the location information extracted for operation **506** of method **500**.

In some embodiments, the set of circuit blocks of the integrated circuit design of method **800** includes at least the set of circuit blocks **704** of FIG. **7** or FIGS. **11A-11C** or set of circuit blocks **904** of FIG. **9A**. In some embodiments, the set of circuit blocks of the integrated circuit design of method **800** includes at least circuit block **602a**, **602b** or **602e** of FIG. **6**.

In some embodiments, the set of heaters of the integrated circuit design of method **800** includes at least set of heaters **706** of FIG. **7**, heater **1160** of FIG. **11C** or set of heaters **906** of FIG. **9A**. In some embodiments, the set of heaters of the integrated circuit design of method **800** includes at least heater **602c** or **602d** of FIG. **6**.

In some embodiments, the power value P of each window (e.g., also referred as “power per window”) of method **800** is determined according to formula 1, and is expressed as:

$$P = NE/PE \quad (1)$$

where NE is the number of elements covered by a first region of the integrated circuit design, and PE is the power consumed by the circuit block or heater within the first region.

A non-limiting exemplary application of formula 1 includes reference to FIG. **9A**.

For example, in some embodiments, CPU **904a** of FIG. **9A** corresponds to the CPU **704a** in layout design **700**, and occupies region **910** in FIG. **9A**. Thus, region **910** occupies or covers 25 total windows, and the power consumed from table **600** for circuit block entry **606a** is 2.5 W, and applying formula 1 to these values results in 2.5 W divided by 25 windows, which is equal to 0.1 W per window. As shown in the FIG. **9A**, region **910** has a power value P that is equal to 0.1 W (shown in key **920**).

In operation **806** of method **800**, the total heat value H_T of each window is determined based on the power value P of each window.

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In some embodiments, the total heat value H_T of each window of method **800** is determined according to formula 2, and is expressed as:

$$H_T = \sum_{j=1}^{m-1} \sum_{i=1}^{n-1} k * P_{ij} / D_{ij} \quad (2)$$

where P_{ij} is the power value of the i th and j th indices, k is a constant based on the material of the circuit block or heater, D_{ij} is the distance between the i th and j th window, m is the number of rows in the array of windows, n is the number of columns in array of windows.

In some embodiments, the total heat value H_T of each window of operation **806** is a sum of the heat generated by the corresponding window and the heat generated by each of the other windows in the integrated circuit design. In some embodiments, the total heat value H_T of each window of operation **806** is a sum of the heat generated by the corresponding window and the heat generated by a number of other windows in the integrated circuit design within a defined distance of the current window. In some embodiments, the value of k is based on the semiconductor material within the heater or the circuit block. In some embodiments, the value of k is equal to 1.

A non-limiting exemplary application of formula 2 includes reference to FIG. 9B. For example, integrated circuit design **900B** corresponds to circuit block **904** of FIG. 9A, but further includes a region **950**, and similar detailed description is omitted. Region **950** includes 4 windows (**950a**, **950b**, **950c** and **950d**), and determining the total heat value H_{T11} of window **950a** includes determining the heat value H_{11} generated by window **950a** and the heat values (H_{21} , H_{12} , H_{22}) generated by each of the other windows (**950b**, **950c**, **950d**) in region **950** of integrated circuit design **900B**. In this example, the region included 4 windows, but other numbers of windows included in the region are within the scope of the present disclosure. Applying formula 2 to the non-limiting example, results in total heat value H_{T11} of window **950a** being equal to $H_{T11} = k * (P_{11} + (P_{12}/D_{12}) + (P_{21}/D_{21}) + (P_{22}/D_{22}))$, where the power values P_{11} , P_{12} , P_{21} and P_{22} are the corresponding power values of the corresponding windows **502a**, **502b**, **502c** and **502d** determined in operation **804** and shown as integrated circuit power map **900A** in FIG. 9A.

In some embodiments, operation **806** is performed for each circuit block in the set of circuit blocks and each heater in the set of heaters in integrated circuit design thereby determining the total heat value of the integrated circuit design.

In some embodiments, by dividing the integrated circuit design into the array of windows, method **800** can accurately determine the heat generated across each window of the integrated circuit design by accounting for not only the heat generated by each window, but also the effect of the heat generated by each of the other windows in the integrated circuit design.

In operation **808** of method **800**, a heat map is populated according to the total heat value of each window thereby generating a heat signature of the integrated circuit design. In some embodiments, operation **808** includes generating a heat map similar to heat map **900C** in FIG. 9C.

In some embodiments, operation **808** includes creating a heat map based on at least the location information of each window in the array of windows and the corresponding total heat value of each corresponding window.

In some embodiments, the heat signature of method **800** includes such as heat signature **930** of FIG. 9C, heat signa-

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tures **1102** and **1106** of FIG. 11A, heat signatures **1122** and **1126** of FIG. 11B, and heat signatures **1122** and **1146** of FIG. 11C.

Method **800** operates to achieve the benefits discussed above with respect to at least system **100** of FIG. 1, wafer **200** of FIG. 2, carrier wafer **300** of FIG. 3 or method **400** of FIG. 4. FIG. 9A is a diagram of a power map **900A**, in accordance with some embodiments.

Power map **900A** corresponds to a power map of integrated circuit design **700**, and similar detailed description is therefore omitted. In some embodiments, power map **900A** is generated by system **1400** while executing operation **804** of FIG. 8.

Power map **900A** includes an integrated circuit design **902** and a power key **920** that illustrates the configured power of each element in the integrated circuit design **902**. In some embodiments, the power key **920** illustrates the configured power of each circuit block of the set of circuit blocks **904** and each heater of the set of heaters **906**.

Integrated circuit design **902** corresponds to integrated circuit design **700**, and similar detailed description is therefore omitted. Integrated circuit design **902** is divided into an array of windows **901**. In some embodiments, dividing integrated circuit design **902** into an array of windows **901** corresponds to operation **802** of FIG. 8.

Integrated circuit design **902** includes a set of circuit blocks **904** and a set of heaters **906**. In some embodiments, set of circuit blocks **904** is corresponding set of circuit blocks **704** of FIG. 7, and set of heaters **906** is corresponding set of heaters **706** of FIG. 7, and similar detailed description is omitted.

Set of circuit blocks **904** includes at least a CPU **904a**, a CPU **904b**, a CPU **904c**, a CPU **904d**, a GPU **904e** or a PLL **904f**. In some embodiments, CPU **904a**, CPU **904b**, CPU **904c**, CPU **904d**, GPU **904e** or PLL **904f** is corresponding CPU **704a**, a CPU **704b**, a CPU **704c**, a CPU **704d**, a GPU **704e** or a PLL **704f** of FIG. 7, and similar detailed description is omitted.

Other numbers of circuits in set of circuit blocks **904** are within the scope of the present disclosure.

Set of heaters **906** includes at least a heater **906a** or a heater **906b**. In some embodiments, heater **906a** or a heater **906b** is corresponding heater **706a** or heater **706b** of FIG. 7, and similar detailed description is omitted. Other numbers of heaters in set of heaters **906** are within the scope of the present disclosure.

Other configurations of power map **900A** are within the scope of the present disclosure.

FIG. 9B is a block diagram of an integrated circuit design **900B**, in accordance with some embodiments.

Integrated circuit design **900B** corresponds to integrated circuit design **902** of FIG. 9A populated with power values P_{11} , P_{12} , P_{21} and P_{22} in corresponding windows **902a**, **902b**, **902c** and **902d**, and similar detailed description is omitted.

In some embodiments, power values P_{11} , P_{12} , P_{21} and P_{22} in corresponding windows **902a**, **902b**, **902c** and **902d** are used by system **1400** while executing operations **804**, **806** or **808** of FIG. 8.

Other configurations of integrated circuit design **900B** are within the scope of the present disclosure.

FIG. 9C is a diagram of a heat map **900C**, in accordance with some embodiments.

Heat map **900C** corresponds to a heat map of power map **900A** or integrated circuit design **700**, and similar detailed description is therefore omitted.

Heat map **900C** corresponds to a heat map of integrated circuit design **700** or power map **900A** of FIG. 9A, and

similar detailed description is therefore omitted. In some embodiments, heat map **900C** is generated by system **1400** while executing operation **808** of FIG. **8**.

Heat map **900C** includes a heat signature **930** of integrated circuit design **902** and a heat key **932** that illustrates the heat of each window in the integrated circuit design **902**. In some embodiments, heat map **900C** is a 2 dimensional contour map of the heat signature **930** of integrated circuit design **902** according to the heat key **932**.

Other configurations of heat map **900C** are within the scope of the present disclosure.

FIG. **10** is a flowchart of a method of modifying an integrated circuit design **1000**, in accordance with some embodiments.

In some embodiments, method **1000** is an embodiment of operation **516** of method **500**, and similar detailed description is omitted. In some embodiments, one or more of operations **1002**, **1004**, **1006**, **1008**, **1010** and **1012** are performed by system **110** or system **1400** of FIG. **14**.

In some embodiments, method **1000** is useable to modify an integrated circuit design such as integrated circuit design **700**, integrated circuit design **902** of FIG. **9A**, integrated circuit design **900B** of FIG. **9B**, integrated circuit design **1112** of FIG. **11A**, integrated circuit design **1132** of FIGS. **11B-11C**, or integrated circuit design **1162** of FIG. **11C**.

In operation **1002** of method **1000**, a determination is made whether the maximum heat values of a first portion of the integrated circuit design exceeds the upper limit of the user defined heat range **HR**.

In some embodiments, the upper limit of the heat range **HR** of method **1000** includes the maximum heat value **Hmax**.

In some embodiments, the maximum heat values of the first portion of the integrated circuit design of method **1000** includes the maximum heat values generated by operation **808** that are in the heat map generated by operation **810**.

In some embodiments, the integrated circuit design of method **1000** includes at least integrated circuit design **700** of FIG. **7**, integrated circuit design **900A** of FIG. **9A**, integrated circuit design **1112** of FIG. **11A**, integrated circuit design **1132** of FIGS. **11B-11C**, or integrated circuit design **1162** of FIG. **11C**.

In some embodiments, the first portion of the integrated circuit design of method **1000** includes any portion of integrated circuit design that satisfies the condition of operation **1002**. In some embodiments, the first portion of the integrated circuit design of method **1000** is also referred to as "hot spot regions."

In some embodiments, if the maximum heat values of the first portion of the integrated circuit design exceeds the upper limit of the user defined heat range **HR**, thus indicating that the heat values **HV** of the heat signature of the integrated circuit design exceed the upper limit of the heat range **HR** specified by the user, then the result of operation **1002** is a "yes", and method **1000** proceeds to operation **1004**.

In some embodiments, if the maximum heat values of the first portion of the integrated circuit design do not exceed the upper limit of the user defined heat range **HR**, thus indicating that the heat values **HV** of the heat signature of the integrated circuit design do not exceed the upper limit of the heat range **HR** specified by the user, then the result of operation **1002** is a "no", and method **1000** proceeds to operation **1006**.

In operation **1004** of method **1000**, the power of circuit blocks or heaters in the first portion of the integrated circuit design is reduced.

In some embodiments, the power of circuit blocks or heaters in the first portion of the integrated circuit design of operation **1004** are reduced according to the set of power entries **604** or the set of power entries **606** contained in table **600** of FIG. **6**.

In some embodiments, if more than one heater or circuit block blocks are in a hot spot region, then operation **1004** comprises reducing at least the power of heaters or the power of circuit blocks that occupy or cover more area in the hot spot region **1114b** (FIG. **11A**). For example, in FIG. **11A**, both heater **706b** and CPU **704d** are in hot spot region **1114b**, and the power of CPU **704d** is reduced by operation **1004** since CPU **704d** occupies more area than heater **706b** in the hot spot region **1114b**.

In some embodiments, if no heater or circuit block are in the hot spot region, then operation **1004** comprises reducing at least the power of heaters or the power of circuit blocks that are separated from the first hot spot region by a lowest distance.

In operation **1006** of method **1000**, a determination is made whether the minimum heat values of a second portion of the integrated circuit design is less than the lower limit of the user heat range.

In some embodiments, the lower limit of the heat range **HR** of method **1000** includes the minimum heat value **Hmin**.

In some embodiments, the minimum heat values of the second portion of the integrated circuit design of method **1000** includes the minimum heat values generated by operation **808** that are in the heat map generated by operation **810**.

In some embodiments, the second portion of the integrated circuit design of method **1000** includes any portion of integrated circuit design that satisfies the condition of operation **1006**. In some embodiments, the second portion of the integrated circuit design of method **1000** is also referred to as "cold spot regions."

In some embodiments, if the minimum heat values of the second portion of the integrated circuit design are less than the lower limit of the user defined heat range **HR**, thus indicating that the heat values **HV** of the heat signature of the integrated circuit design are less than the lower limit of the heat range **HR** specified by the user, then the result of operation **1006** is a "yes", and method **1000** proceeds to operation **1008**.

In some embodiments, if the minimum heat values of the second portion of the integrated circuit design are not less than the lower limit of the user defined heat range **HR**, thus indicating that the heat values **HV** of the heat signature of the integrated circuit design are not less than the lower limit of the heat range **HR** specified by the user, then the result of operation **1006** is a "no", and method **1000** proceeds to operation **508**.

In operation **1008** of method **1000**, a determination is made whether the power of circuit blocks or heaters in the second portion of the integrated circuit design is at maximum power.

In some embodiments, operation **1008** includes determining whether the power of circuit blocks or heaters in the "cold spot regions" of the integrated circuit design are at maximum power.

In some embodiments, if the power of circuit blocks or heaters in the second portion of the integrated circuit design are at maximum power, thus indicating that the power values of circuit blocks or heaters in the second portion of the integrated circuit design cannot be further increased, then the result of operation **1008** is a "yes", and method **1000** proceeds to operation **1010**.

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In some embodiments, if the power of circuit blocks or heaters in the second portion of the integrated circuit design are not at maximum power, thus indicating that the power values of circuit blocks or heaters in the second portion of the integrated circuit design can be further increased, then the result of operation 1008 is a “no”, and method 1000 proceeds to operation 1012.

In operation 1010 of method 1000, at least a heater is inserted in the second portion of the integrated circuit design. In some embodiments, the inserted heater of method 1000 includes heater 1160 of FIG. 11C.

In some embodiments, operation 1010 includes inserting at least a heater in first cold spot regions of the integrated circuit design.

In operation 1012 of method 1000, power of circuit blocks or heaters in the second portion of the integrated circuit design are increased. In some embodiments, a heater with power that is increased by operation 1012 includes heater 706a in FIG. 11B.

Method 1000 operates to achieve the benefits discussed above with respect to at least system 100 of FIG. 1, wafer 200 of FIG. 2, carrier wafer 300 of FIG. 3, method 400 of FIG. 4 or FIGS. 5-11C.

FIG. 11A is a diagram 1100A of power maps and an integrated circuit design, in accordance with some embodiments.

Diagram 1100A includes heat map 1102, heat map 1106 and integrated circuit design 1112.

In some embodiments, integrated circuit design 1112 corresponds to integrated circuit design 700, and heat map 1102 corresponds to heat map 900C of FIG. 9C, and similar detailed description is therefore omitted.

Heat map 1102 corresponds to a heat map of integrated circuit design 1112, and similar detailed description is therefore omitted. In some embodiments, heat maps 1102 and 1106 are generated by system 1400 while executing operation 804 of FIG. 8.

Heat map 1106 corresponds to a heat map of integrated circuit design 1112 after execution of operation 1004 of FIG. 10, and similar detailed description is therefore omitted.

Heat map 1102 includes hot spot region 1104a and hot spot region 1104b. Hot spot region 1104a corresponds to region 1114a of integrated circuit design 1112 and hot spot region 1104b corresponds to region 1114b of integrated circuit design 1112.

Region 1114a of integrated circuit design 1112 covers CPU 704c. Region 1114b of integrated circuit design 1112 covers part of CPU 704d and heater 706b.

Heat map 1106 includes hot spot region 1108a and hot spot region 1108b. Hot spot region 1108a corresponds to hot spot region 1104a after execution of operation 1004, and hot spot region 1108b corresponds to hot spot region 1104b after execution of operation 1004.

For example, in some embodiments, during execution of operation 1004, CPU 704c is in region 1114a that corresponds to hot spot region 1104a, and the power of CPU 704c is reduced by operation 1004 thereby generating hot spot region 1108a of heat map 1106.

For example, in some embodiments, during execution of operation 1004, both heater 706b and CPU 704d are in region 1114b that corresponds to hot spot region 1104b, and the power of CPU 704d is reduced by operation 1004 since CPU 704d occupies more area than heater 706b in region 1114b thereby generating hot spot region 1108b of heat map 1106.

Other configurations of diagram 1100A are within the scope of the present disclosure.

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FIG. 11B is a diagram 1100B of power maps and an integrated circuit design, in accordance with some embodiments.

Diagram 1100B includes heat map 1122, heat map 1126 and integrated circuit design 1132.

In some embodiments, integrated circuit design 1132 corresponds to integrated circuit design 700, and heat map 1122 corresponds to heat map 900C of FIG. 9C, and similar detailed description is therefore omitted.

Heat map 1122 corresponds to a heat map of integrated circuit design 1132, and similar detailed description is therefore omitted. In some embodiments, heat maps 1122 and 1126 are generated by system 1400 while executing operation 804 of FIG. 8.

Heat map 1126 corresponds to a heat map of integrated circuit design 1132 after execution of operation 1012 of FIG. 10, and similar detailed description is therefore omitted.

Heat map 1122 includes cold spot region 1124a. Cold spot region 1124a corresponds to region 1134a of integrated circuit design 1132.

Region 1134a of integrated circuit design 1132 corresponds to CPU 704c.

Heat map 1126 includes cold spot region 1128a. Cold spot region 1128a corresponds to cold spot region 1124a after execution of operation 1012. For example, in some embodiments, during execution of operation 1012, heater 706a is in region 1134a that corresponds to cold spot region 1124a, and the power of heater 706a is increased by operation 1012 thereby generating cold spot region 1128a of heat map 1126.

Other configurations of diagram 1100B are within the scope of the present disclosure.

FIG. 11C is a diagram 1100C of power maps and an integrated circuit design, in accordance with some embodiments.

Diagram 1100C includes heat map 1122, heat map 1146, integrated circuit design 1132 and integrated circuit design 1162.

In some embodiments, integrated circuit design 1132 corresponds to integrated circuit design 700, and heat map 1122 corresponds to heat map 900C of FIG. 9C, and similar detailed description is therefore omitted.

Heat map 1122 corresponds to a heat map of integrated circuit design 1132, and similar detailed description is therefore omitted. In some embodiments, heat maps 1122 and 1146 are generated by system 1400 while executing operation 804 of FIG. 8.

Heat map 1146 corresponds to a heat map of integrated circuit design 1162, and the integrated circuit design 1162 and corresponding heat map 1146 are generated after execution of operation 1010 of FIG. 10, and similar detailed description is therefore omitted.

Heat map 1122 includes cold spot region 1144a. Cold spot region 1144a corresponds to region 1154a of integrated circuit design 1132.

Region 1154a of integrated circuit design 1132 does not include a circuit block or heater in integrated circuit design 1132.

Heat map 1146 includes cold spot region 1148a. Cold spot region 1148a corresponds to cold spot region 1144a after execution of operation 1010. For example, in some embodiments, during execution of operation 1010, no heater is in region 1154a of integrated circuit design 1132, so system 1400 inserts new heater 1160 into region 1164a of integrated circuit design 1132, and region 1164a corresponds to cold spot region 1148a of heat map 1146.

Other configurations of diagram 1100C are within the scope of the present disclosure.

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FIG. 12 is a flowchart of a method 1200 of simultaneously performing a burn-in test of the integrated circuit and an automated test of the integrated circuit, in accordance with some embodiments.

In some embodiments, method 1200 is an embodiment of operation 408 of method 400, and similar detailed description is omitted. In some embodiments, one or more of operations 1202, 1204 and 1206 are performed by system 100 or system 1400 of FIG. 14. In some embodiments, method 1200 is useable to simultaneously perform a burn-in test of the integrated circuit and an automated test of the integrated circuit.

In operation 1202 of method 1200, a set of circuit blocks and a first set of heaters are configured as a first set of heat sources for the burn-in test of the integrated circuit thereby generating a first heat signature of the integrated circuit.

In some embodiments, operation 1202 comprises turning on the set of circuit blocks and the first set of heaters according to simulated design power levels corresponding to the integrated circuit design thereby generating the first heat signature.

In some embodiments, the integrated circuit of method 1200 includes at least wafer 102, wafer 200, plurality of integrated circuits 201 or integrated circuit 203.

In some embodiments, the integrated circuit design of method 1200 includes at least integrated circuit design 700 of FIG. 7, integrated circuit design 902 of FIG. 9A, integrated circuit design 1112 of FIG. 11A, integrated circuit design 1132 of FIGS. 11B-11C, or integrated circuit design 1162 of FIG. 11C.

In some embodiments, the set of circuit blocks of the integrated circuit of method 1200 includes at least the set of circuit blocks 204 of FIG. 2.

In some embodiments, the first set of heaters of the integrated circuit of method 1200 includes at least set of heaters 206 of FIG. 2.

In some embodiments, the simulated design power levels of method 1200 includes at least one or more power entries 604 or 606 contained in table 600.

In operation 1204 of method 1200, the integrated circuit is placed on a carrier wafer. In some embodiments, the carrier wafer of method 1200 includes carrier wafer 106 of FIG. 1 and carrier wafer 300 of FIG. 3.

In operation 1206 of method 1200, at least a portion of the carrier wafer is configured as a second set of heat sources for the burn-in test of the integrated circuit thereby generating a second heat signature of the integrated circuit.

In some embodiments, the at least a portion of the carrier wafer of method 1200 includes at least region 302 of FIG. 3, one or more of the plurality of integrated circuit dies 301, integrated circuit die 303 or at least a die of the set of circuit dies 304.

In some embodiments, operation 1206 comprises turning on the second set of heaters thereby generating the second heat signature corresponding to the second set of heat sources for the burn-in test of the integrated circuit.

In some embodiments, the at least second set of heaters of method 1200 includes the heater chip 108 of FIG. 1, at least a heater of the set of heaters 306, heater 1300 of FIG. 13 or at least a portion of interconnect 1302a of FIG. 13 or interconnect 1302b of FIG. 13.

In some embodiments, the first heat signature of the integrated circuit of method 1200 corresponds to the burn-in heat distribution of the integrated circuit, and the heaters in the second set of heaters are turned off.

In some embodiments, the first heat signature of the integrated circuit and the second heat signature of method

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1200 correspond to the burn-in heat distribution of the integrated circuit, and at least one heater in the second set of heaters is turned on.

Method 1200 operates to achieve the benefits discussed above with respect to at least system 100 of FIG. 1, wafer 200 of FIG. 2, carrier wafer 300 of FIG. 3, method 400 of FIG. 4 or FIGS. 5-11C.

FIG. 13 is a cross-sectional view of a heater 1300, in accordance with some embodiments. In some embodiments, heater 1300 is an embodiment of heater chip 108 in FIG. 1 or a heater of the set of heaters 306 in FIG. 3, and similar detailed description is omitted.

Heater 1300 is useable as a heater in heater chip 108 in FIG. 1 or a heater of the set of heaters 306 in FIG. 3.

Heater 1300 includes an interconnect 1302a and an interconnect 1302b in one or more layers of a die 1304. In some embodiments, die 1304 corresponds to one or more dies in set of dies 304 of FIG. 3, and similar detailed description is omitted.

Interconnects 1302a and 1302b each include one or more conductive features, such as conductive lines (not labelled), vias (not labelled), or conductive pads (not labelled), formed in an insulating material 1310. In some embodiments, the one or more conductive features, such as the conductive lines (not labelled), the vias (not labelled), or the conductive pads (not labelled), formed in insulating material 1310 of interconnect structure 1302a and 1302b is referred to as one or more redistribution layers (RDL) of die 1304. The routings of the conductive features shown in FIG. 13 are merely examples. Other configurations, arrangements and materials of the conductive features of interconnect structures 1302a and 1302b are within the contemplated scope of the present disclosure. Other configurations, arrangements and materials of interconnect structures 1302a and 1302b are within the contemplated scope of the present disclosure.

In some embodiments, interconnects 1302a and 1302b are configured to generate heat and function as 2 different stages of corresponding heat sources when electrical current passes through the corresponding interconnects 1302a and 1302b. In some embodiments, the amount of heat generated by interconnects 1302a and 1302b can be adjusted by changing the amount of current passing through corresponding interconnects 1302a and 1302b. In some embodiments, the amount of heat generated by interconnects 1302a and 1302b can be adjusted by selecting a different number of stages of interconnects 1302a and 1302b.

In some embodiments, at least interconnect structure 1302a or 1302b is made of conductive materials, such as copper, copper alloy, aluminum, alloys or combinations thereof. In some embodiments, other applicable materials are used. In some embodiments, at least interconnect structure 1302a or 1302b include other conductive materials, such as tungsten (W), Cu, Al, or AlCu. In some embodiments, insulating material 1310 is made of silicon oxide. In some embodiments, insulating material 1310 includes multiple dielectric layers of dielectric materials. One or more of the multiple dielectric layers are made of low dielectric constant (low-k) materials.

FIG. 13 shows two interconnects or stages, but other numbers of stages of interconnects 1302a and 1302b and corresponding current values passing through interconnects 1302a and 1302b are within the scope of the present disclosure.

Heater 1300 further includes an under bump metallurgy (UBM) layer 1312 on a surface of the interconnect structure 1302a and 1302b. In some embodiments, the UBM layer includes one or more conductive portions 1312a,

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1312b, . . . , **1312f**/where **f** is an integer corresponding to the number of conductive portions in the UBM layer **1312**. In some embodiments, UBM layer **1312** is formed on the surface of the interconnect structure **1302a** and **1302b**. In some embodiments, UBM layer **1312** is formed on a metal pad (not shown). In some embodiments, UBM layer **1312** includes an adhesion layer and/or a wetting layer. In some embodiments, UBM layer **1312** includes at least a copper seed layer. In some embodiments, UBM layer **1312** includes titanium (Ti), titanium nitride (TiN), tantalum nitride (TaN), tantalum (Ta), or the like. Other configurations, arrangements and materials of UBM layer **1312** are within the contemplated scope of the present disclosure.

Integrated circuit **1300** further includes a set of solder bumps **1314** on the UBM layer **1312**.

The set of solder bumps **1314** includes one or more solder bumps **1314a**, **1314b**, . . . , **1314f**, where **f** is an integer corresponding to the number of solder bumps in the set of solder bumps **1314**. In some embodiments, the set of solder bumps **1314** is formed over UBM layer **1312**. In some embodiments, one or more solder bumps **1314a**, **1314b**, . . . , **1314f**/of the set of solder bumps **1314** includes a conductive material having a low resistivity, such as solder or a solder alloy. In some embodiments, a solder alloy includes Sn, Pb, Ag, Cu, Ni, Bi, or combinations thereof. Other configurations, arrangements and materials of the set of solder bumps **1314** are within the contemplated scope of the present disclosure.

In some embodiments, the set of solder bumps **1314** is electrically connected to one or more other package structures (not shown) by the set of bumps **1314**.

Other configurations, arrangements and materials of heater **1300** are within the contemplated scope of the present disclosure.

FIG. **14** is a schematic view of a system **1400** for designing an IC layout design, simulating an IC design, and manufacturing an IC circuit in accordance with some embodiments. In some embodiments, system **1400** generates or places one or more IC layout designs described herein, tests one or more IC designs based on the IC layout design and manufactures an IC based on the layout design. In some embodiments, system **1400** is system **110** in FIG. **1**, and similar detailed description is therefore omitted.

System **1400** includes a hardware processor **1402** and a non-transitory, computer readable storage medium **1404** (e.g., memory **1404**) encoded with, i.e., storing, the computer program code **1406**, i.e., a set of executable instructions **1406**. In some embodiments, computer readable storage medium **1404** is configured for interfacing with manufacturing machines for producing the integrated circuit. In some embodiments, computer readable storage medium **1404** is configured for interfacing with wafer **102**, test circuit board **104**, carrier wafer **106**, heater chip **108** for testing the integrated circuit. In some embodiments, computer readable storage medium **1404** is configured to generate and simulate the integrated circuit design.

The processor **1402** is electrically coupled to the computer readable storage medium **1404** via a bus **1408**. The processor **1402** is also electrically coupled to an I/O interface **1410** by bus **1408**. A network interface **1412** is also electrically connected to the processor **1402** via bus **1408**. Network interface **1412** is connected to a network **1414**, so that processor **1402** and computer readable storage medium **1404** are capable of connecting to external elements via network **1414**. The processor **1402** is configured to execute the computer program code **1406** encoded in the computer readable storage medium **1404** in order to cause system

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1400 to be usable for performing a portion or all of the operations as described in at least method **400**, **500**, **800**, **1000** or **1200**.

In some embodiments, the processor **1402** is a central processing unit (CPU), a multi-processor, a distributed processing system, an application specific integrated circuit (ASIC), and/or a suitable processing unit.

In some embodiments, the computer readable storage medium **1404** is an electronic, magnetic, optical, electro-magnetic, infrared, and/or a semiconductor system (or apparatus or device). For example, the computer readable storage medium **1404** includes a semiconductor or solid-state memory, a magnetic tape, a removable computer diskette, a random access memory (RAM), a read-only memory (ROM), a rigid magnetic disk, and/or an optical disk. In some embodiments using optical disks, the computer readable storage medium **1404** includes a compact disk-read only memory (CD-ROM), a compact disk-read/write (CD-R/W), and/or a digital video disc (DVD).

In some embodiments, the storage medium **1404** stores the computer program code **1406** configured to cause system **1400** to perform at least method **400**, **500**, **800**, **1000** or **1200**. In some embodiments, the storage medium **1404** also stores information needed for performing at least method **400**, **500**, **800**, **1000** or **1200** as well as information generated during performing at least method **400**, **500**, **800**, **1000** or **1200**, such as IC design **1416**, user interface **1418**, IC design simulation **1422**, simulation parameters **1424**, test parameters **1426**, lookup table **1428** and maps **1430**, and/or a set of executable instructions to perform the operation of at least method **400**, **500**, **800**, **1000** or **1200**. In some embodiments, IC design **1416** comprises one or more of layout patterns of wafer **200**, integrated circuit design **700**, **900A-900B**, **1112**, **1132** or **1162**.

In some embodiments, IC design **1416** includes at least integrated circuit design **700** of FIG. **7**, integrated circuit design **902** of FIG. **9A**, integrated circuit design **1112** of FIG. **11A**, integrated circuit design **1132** of FIGS. **11B-11C**, or integrated circuit design **1162** of FIG. **11C**.

In some embodiments, IC design simulation **1422** includes computer code configured to perform one or more portions of at least method **400**, **500**, **800**, **1000** or **1200**.

In some embodiments, simulation parameters **1424** includes at least a number of iterations of method **500** as defined by one or more users, one or more heat parameters or one or more power parameters of one or more of FIGS. **1-13**. In some embodiments, one or more heat parameters include at least heat distribution, one or more heat ranges from user(s), one or more heat signatures, one or more heat values, one or more total heat values in one or more of FIGS. **1-13**. In some embodiments, one or more power parameters include at least power information from table **600**, power information or one or more power values of each window of array of windows in one or more of FIGS. **1-13**.

In some embodiments, test parameters **1426** includes at least burn-in testing parameters and ATE testing parameters in one or more of FIGS. **1-13**.

In some embodiments, lookup table **1428** includes at least table **600**, power information, or location information in one or more of FIGS. **1-13**. In some embodiments, lookup table **1428** includes the design file of method **500**.

In some embodiments, maps **1430** includes at least one or more heat maps, one or more power maps, one or more hot spots, one or more cold spots or one or more array of windows in one or more of FIGS. **1-13**.

In some embodiments, the storage medium **1404** stores instructions (e.g., computer program code **1406**) for inter-

facing with manufacturing machines. The instructions (e.g., computer program code **1406**) enable processor **1402** to generate manufacturing instructions readable by the manufacturing machines to effectively implement at least method **400**, **500**, **800**, **1000** or **1200** during a manufacturing process.

System **1400** includes I/O interface **1410**. I/O interface **1410** is coupled to external circuitry. In some embodiments, I/O interface **1410** includes a keyboard, keypad, mouse, trackball, trackpad, and/or cursor direction keys for communicating information and commands to processor **1402**.

System **1400** also includes network interface **1412** coupled to the processor **1402**. Network interface **1412** allows system **1400** to communicate with network **1414**, to which one or more other computer systems are connected. Network interface **1412** includes wireless network interfaces such as BLUETOOTH, WIFI, WIMAX, GPRS, or WCDMA; or wired network interface such as ETHERNET, USB, or IEEE-1494. In some embodiments, at least method **400**, **500**, **800**, **1000** or **1200** is implemented in two or more systems **1400**, and information such as IC design, user interface, IC design simulation, simulation parameters, test parameters, lookup table and maps are exchanged between different systems **1400** by network **1414**.

System **1400** is configured to receive information related to an IC or layout design through I/O interface **1410** or network interface **1412**. The information is transferred to processor **1402** by bus **1408** to determine an IC design for producing at least wafer **102**, wafer **200**, plurality of integrated circuits **201** or integrated circuit **203**. The IC design is then stored in computer readable medium **1404** as IC design **1416**. System **1400** is configured to receive information related to a user interface through I/O interface **1410** or network interface **1412**. The information is stored in computer readable medium **1404** as user interface **1418**. System **1400** is configured to receive information related to an IC design simulation through I/O interface **1410** or network interface **1412**. The information is stored in computer readable medium **1404** as IC design simulation **1422**. System **1400** is configured to receive information related to simulation parameters through I/O interface **1410** or network interface **1412**. The information is stored in computer readable medium **1404** as simulation parameters **1424**. System **1400** is configured to receive information related to test parameters through I/O interface **1410** or network interface **1412**. The information is stored in computer readable medium **1404** as test parameters **1426**. System **1400** is configured to receive information related to lookup table through I/O interface **1410** or network interface **1412**. The information is stored in computer readable medium **1404** as lookup table **1428**. System **1400** is configured to receive information related to maps through I/O interface **1410** or network interface **1412**. The information is stored in computer readable medium **1404** as maps **1430**.

In some embodiments, at least portions of one or more of method **400**, **500**, **800**, **1000** or **1200** is implemented as a standalone software application for execution by a processor. In some embodiments, at least portions of one or more of method **400**, **500**, **800**, **1000** or **1200** is implemented as a software application that is a part of an additional software application. In some embodiments, at least portions of one or more of method **400**, **500**, **800**, **1000** or **1200** is implemented as a plug-in to a software application. In some embodiments, at least portions of one or more of method **400**, **500**, **800**, **1000** or **1200** is implemented as a software application that is a portion of an EDA tool. In some

400, **500**, **800**, **1000** or **1200** is implemented as a software application that is used by an EDA tool. In some embodiments, the EDA tool is used to generate a layout design of an integrated circuit device, generate and simulate of an integrated circuit design of an IC device or perform testing of the IC device. In some embodiments, the layout design, IC design and simulation and testing parameters are stored on a non-transitory computer readable medium. In some embodiments, the layout is generated using a tool such as VIRTUOSO® available from CADENCE DESIGN SYSTEMS, Inc., or another suitable layout generating tool. In some embodiments, the layout is generated based on a netlist which is created based on the schematic design. In some embodiments, at least portions of one or more of method **400**, **500**, **800**, **1000** or **1200** is implemented by a manufacturing device to manufacture an integrated circuit using a set of masks manufactured based on one or more layout designs generated by system **1400**. In some embodiments, system **1400** is a manufacturing device to manufacture an integrated circuit using a set of masks manufactured based on one or more layout designs of the present disclosure.

In some embodiments, system **1400** of FIG. **14** operates to achieve the benefits discussed above with respect to at least system **100** of FIG. **1**, wafer **200** of FIG. **2**, carrier wafer **300** of FIG. **3**, method **400** of FIG. **4** or FIGS. **5-13**.

FIG. **15** is a block diagram of an integrated circuit (IC) manufacturing system **1500**, and an IC manufacturing flow associated therewith, in accordance with at least one embodiment of the present disclosure. In some embodiments, based on a layout diagram, at least one of (A) one or more semiconductor masks or (B) at least one component in a layer of a semiconductor integrated circuit is fabricated using manufacturing system **1500**.

In FIG. **15**, IC manufacturing system **1500** (hereinafter “system **1500**”) includes entities, such as a design house **1520**, a mask house **1530**, and an IC manufacturer/fabricator (“fab”) **1540**, that interact with one another in the design, development, and manufacturing cycles and/or services related to manufacturing an IC device **1560**. The entities in system **1500** are connected by a communications network. In some embodiments, the communications network is a single network. In some embodiments, the communications network is a variety of different networks, such as an intranet and the Internet. The communications network includes wired and/or wireless communication channels. Each entity interacts with one or more of the other entities and provides services to and/or receives services from one or more of the other entities. In some embodiments, one or more of design house **1520**, mask house **1530**, and IC fab **1540** is owned by a single larger company. In some embodiments, one or more of design house **1520**, mask house **1530**, and IC fab **1540** coexist in a common facility and use common resources.

Design house (or design team) **1520** generates an IC design layout **1522**. IC design layout **1522** includes various geometrical patterns designed for an IC device **1560**. The geometrical patterns correspond to patterns of metal, oxide, or semiconductor layers that make up the various components of IC device **1560** to be fabricated. The various layers combine to form various IC features. For example, a portion of IC design layout **1522** includes various IC features, such as an active region, gate electrode, source electrode and drain electrode, metal lines or vias of an interlayer interconnection, and openings for bonding pads, to be formed in a semiconductor substrate (such as a silicon wafer) and various material layers disposed on the semiconductor substrate. Design house **1520** implements a proper design procedure to form IC design layout **1522**. The design

procedure includes one or more of logic design, physical design or place and route. IC design layout **1522** is presented in one or more data files having information of the geometrical patterns. For example, IC design layout **1522** can be expressed in a GDSII file format or DFII file format. In some embodiments, IC design layout **1522** includes at least integrated circuit design **700** of FIG. 7, integrated circuit design **902** of FIG. 9A, integrated circuit design **1112** of FIG. 11A, integrated circuit design **1132** of FIGS. 11B-11C, or integrated circuit design **1162** of FIG. 11C.

Mask house **1530** includes data preparation **1532** and mask fabrication **1534**. Mask house **1530** uses IC design layout **1522** to manufacture one or more masks **1545** to be used for fabricating the various layers of IC device **1560** according to IC design layout **1522**. Mask house **1530** performs mask data preparation **1532**, where IC design layout **1522** is translated into a representative data file ("RDF"). Mask data preparation **1532** provides the RDF to mask fabrication **1534**. Mask fabrication **1534** includes a mask writer. A mask writer converts the RDF to an image on a substrate, such as a mask (reticle) **1545** or a semiconductor wafer **1542**. The design layout **1522** is manipulated by mask data preparation **1532** to comply with particular characteristics of the mask writer and/or requirements of IC fab **1540**. In FIG. 15, mask data preparation **1532** and mask fabrication **1534** are illustrated as separate elements. In some embodiments, mask data preparation **1532** and mask fabrication **1534** can be collectively referred to as mask data preparation.

In some embodiments, mask data preparation **1532** includes optical proximity correction (OPC) which uses lithography enhancement techniques to compensate for image errors, such as those that can arise from diffraction, interference, other process effects and the like. OPC adjusts IC design layout **1522**. In some embodiments, mask data preparation **1532** includes further resolution enhancement techniques (RET), such as off-axis illumination, sub-resolution assist features, phase-shifting masks, other suitable techniques, and the like or combinations thereof. In some embodiments, inverse lithography technology (ILT) is also used, which treats OPC as an inverse imaging problem.

In some embodiments, mask data preparation **1532** includes a mask rule checker (MRC) that checks the IC design layout that has undergone processes in OPC with a set of mask creation rules which contain certain geometric and/or connectivity restrictions to ensure sufficient margins, to account for variability in semiconductor manufacturing processes, and the like. In some embodiments, the MRC modifies the IC design layout to compensate for limitations during mask fabrication **1534**, which may undo part of the modifications performed by OPC in order to meet mask creation rules.

In some embodiments, mask data preparation **1532** includes lithography process checking (LPC) that simulates processing that will be implemented by IC fab **1540** to fabricate IC device **1560**. LPC simulates this processing based on IC design layout **1522** to create a simulated manufactured device, such as IC device **1560**. The processing parameters in LPC simulation can include parameters associated with various processes of the IC manufacturing cycle, parameters associated with tools used for manufacturing the IC, and/or other aspects of the manufacturing process. LPC takes into account various factors, such as aerial image contrast, depth of focus ("DOF"), mask error enhancement factor ("MEEF"), other suitable factors, and the like or combinations thereof. In some embodiments, after a simulated manufactured device has been created by

LPC, if the simulated device is not close enough in shape to satisfy design rules, OPC and/or MRC are repeated to further refine IC design layout **1522**.

It should be understood that the above description of mask data preparation **1532** has been simplified for the purposes of clarity. In some embodiments, data preparation **1532** includes additional features such as a logic operation (LOP) to modify the IC design layout according to manufacturing rules. Additionally, the processes applied to IC design layout **1522** during data preparation **1532** may be executed in a variety of different orders.

After mask data preparation **1532** and during mask fabrication **1534**, a mask **1545** or a group of masks **1545** are fabricated based on the modified IC design layout **1522**. In some embodiments, mask fabrication **1534** includes performing one or more lithographic exposures based on IC design layout **1522**. In some embodiments, an electron-beam (e-beam) or a mechanism of multiple e-beams is used to form a pattern on a mask (photomask or reticle) **1545** based on the modified IC design layout **1522**. The mask **1545** can be formed in various technologies. In some embodiments, the mask **1545** is formed using binary technology. In some embodiments, a mask pattern includes opaque regions and transparent regions. A radiation beam, such as an ultraviolet (UV) beam, used to expose the image sensitive material layer (e.g., photoresist) which has been coated on a wafer, is blocked by the opaque region and transmits through the transparent regions. In one example, a binary version of mask **1545** includes a transparent substrate (e.g., fused quartz) and an opaque material (e.g., chromium) coated in the opaque regions of the binary mask. In another example, the mask **1545** is formed using a phase shift technology. In the phase shift mask (PSM) version of mask **1545**, various features in the pattern formed on the mask are configured to have proper phase difference to enhance the resolution and imaging quality. In various examples, the phase shift mask can be attenuated PSM or alternating PSM. The mask(s) generated by mask fabrication **1534** is used in a variety of processes. For example, such a mask(s) is used in an ion implantation process to form various doped regions in the semiconductor wafer, in an etching process to form various etching regions in the semiconductor wafer, and/or in other suitable processes.

IC fab **1540** is an IC fabrication entity that includes one or more manufacturing facilities for the fabrication of a variety of different IC products. In some embodiments, IC Fab **1540** is a semiconductor foundry. For example, there may be a manufacturing facility for the front end fabrication of a plurality of IC products (front-end-of-line (FEOL) fabrication), while a second manufacturing facility may provide the back end fabrication for the interconnection and packaging of the IC products (back-end-of-line (BEOL) fabrication), and a third manufacturing facility may provide other services for the foundry entity.

IC fab **1540** includes wafer fabrication tools **1552** (hereinafter "fabrication tools **1552**") configured to execute various manufacturing operations on semiconductor wafer **1542** such that IC device **1560** is fabricated in accordance with the mask(s), e.g., mask **1545**. In various embodiments, fabrication tools **1552** include one or more of a wafer stepper, an ion implanter, a photoresist coater, a process chamber, e.g., a CVD chamber or LPCVD furnace, a CMP system, a plasma etch system, a wafer cleaning system, or other manufacturing equipment capable of performing one or more suitable manufacturing processes as discussed herein.

IC fab **1540** uses mask(s) **1545** fabricated by mask house **1530** to fabricate IC device **1560**. Thus, IC fab **1540** at least

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indirectly uses IC design layout **1522** to fabricate IC device **1560**. In some embodiments, a semiconductor wafer **1542** is fabricated by IC fab **1540** using mask(s) **1545** to form IC device **1560**. In some embodiments, the IC fabrication includes performing one or more lithographic exposures based at least indirectly on IC design layout **1522**. Semiconductor wafer **1542** includes a silicon substrate or other proper substrate having material layers formed thereon. Semiconductor wafer **1542** further includes one or more of various doped regions, dielectric features, multilevel interconnects, and the like (formed at subsequent manufacturing steps).

In some embodiments, IC device **1560** includes at least wafer **102**, wafer **200**, plurality of integrated circuits **201** or integrated circuit **203**.

System **1500** is shown as having design house **1520**, mask house **1530** or IC fab **1540** as separate components or entities. However, it is understood that one or more of design house **1520**, mask house **1530** or IC fab **1540** are part of the same component or entity.

Details regarding an integrated circuit (IC) manufacturing system (e.g., system **1500** of FIG. **15**), and an IC manufacturing flow associated therewith are found, e.g., in U.S. Pat. No. 9,256,709, granted Feb. 9, 2016, U.S. Pre-Grant Publication No. 20150278429, published Oct. 1, 2015, U.S. Pre-Grant Publication No. 20140040838, published Feb. 6, 2014, and U.S. Pat. No. 7,260,442, granted Aug. 21, 2007, the entireties of each of which are hereby incorporated by reference.

It will be readily seen by one of ordinary skill in the art that one or more of the disclosed embodiments fulfill one or more of the advantages set forth above. After reading the foregoing specification, one of ordinary skill will be able to affect various changes, substitutions of equivalents and various other embodiments as broadly disclosed herein. It is therefore intended that the protection granted hereon be limited only by the definition contained in the appended claims and equivalents thereof.

One aspect of this description relates to a method of testing an integrated circuit on a test circuit board. In some embodiments, the method includes performing, by a processor, a simulation of a first heat distribution throughout an integrated circuit design, the integrated circuit design corresponding to the integrated circuit. In some embodiments, the performing the simulation includes determining a heat signature of the integrated circuit design from at least configured power information or location information for each circuit block of the set of circuit blocks or each heater of the set of heaters included in the integrated circuit design, the heat signature including heat values distributed throughout the integrated circuit design. In some embodiments, the performing the simulation further includes simultaneously performing a burn-in test of the integrated circuit and an automated test of the integrated circuit, the integrated circuit being coupled to the test circuit board, wherein the burn-in test has at least one of a minimum burn-in temperature of the integrated circuit or a burn-in heat distribution across the integrated circuit.

Another aspect of this description relates to a method of testing an integrated circuit on a test circuit board. In some embodiments, the method includes performing, by a processor, a simulation of a first heat distribution throughout an integrated circuit design, the integrated circuit design including at least a set of circuit blocks. In some embodiments, the performing the simulation includes obtaining configured power information for at least each circuit block of the set of circuit blocks in the integrated circuit design. In

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some embodiments, the performing the simulation further includes determining a heat signature of the integrated circuit design from at least the configured power information for each circuit block of the set of circuit blocks included in the integrated circuit design, the heat signature including heat values distributed throughout the integrated circuit design. In some embodiments, the performing the simulation further includes modifying the integrated circuit design in response to at least determining that the heat values of the heat signature of the integrated circuit design are not within a heat range. In some embodiments, the method further includes manufacturing the integrated circuit according to the integrated circuit design. In some embodiments, the method further includes simultaneously performing a burn-in test of the integrated circuit and an automated test of the integrated circuit, the integrated circuit being configured to operate according to simulated design power levels and being coupled to the test circuit board. In some embodiments, the burn-in test has a minimum burn-in temperature of the integrated circuit and a burn-in heat distribution across the integrated circuit.

Still another aspect of this description relates to a testing system. In some embodiments, the testing system includes an integrated circuit, a carrier wafer coupled to at least the integrated circuit, and a first system electrically coupled to the integrated circuit. In some embodiments, the first system includes a non-transitory computer readable medium configured to store executable instructions, and a processor coupled to the non-transitory computer readable medium. In some embodiments, the processor is configured to execute the executable instructions for performing a simulation of a first heat distribution throughout an integrated circuit design, the integrated circuit design corresponding to the integrated circuit. In some embodiments, the testing system is configured to simultaneously perform a burn-in test of the integrated circuit and an automated test of the integrated circuit. In some embodiments, the burn-in test has a minimum burn-in temperature of the integrated circuit and a burn-in heat distribution across the integrated circuit. In some embodiments, the testing system configured to simultaneously perform the burn-in test of the integrated circuit and the automated test of the integrated circuit includes configuring a set of circuit blocks and a first set of heaters as a first set of heat sources for the burn-in test of the integrated circuit thereby generating a first heat signature of the integrated circuit.

The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. A method of testing an integrated circuit on a test circuit board, the integrated circuit including at least a set of circuit blocks or a first set of heaters, the method comprising:
 - performing, by a processor, a simulation of a first heat distribution throughout an integrated circuit design, the

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integrated circuit design corresponding to the integrated circuit, the performing the simulation comprising:

determining a heat signature of the integrated circuit design from at least configured power information or location information for each circuit block of the set of circuit blocks or each heater of the set of heaters included in the integrated circuit design, the heat signature including heat values distributed throughout the integrated circuit design; and

simultaneously performing a burn-in test of the integrated circuit and an automated test of the integrated circuit, the integrated circuit being coupled to the test circuit board, wherein the burn-in test has at least one of a minimum burn-in temperature of the integrated circuit or a burn-in heat distribution across the integrated circuit.

2. The method of claim 1, wherein the determining the heat signature of the integrated circuit design from at least the configured power information or the location information for each circuit block of the set of circuit blocks or each heater of the set of heaters included in the integrated circuit design comprises:

dividing the integrated circuit design into a grid of windows based on a window size.

3. The method of claim 2, wherein the determining the heat signature of the integrated circuit design from at least the configured power information or the location information for each circuit block of the set of circuit blocks or each heater of the set of heaters included in the integrated circuit design further comprises:

determining a power value of each window based on at least the power information or the location information of each circuit block of the set of circuit blocks.

4. The method of claim 3, wherein the determining the power value of each window based on at least the power information or the location information of each circuit block of the set of circuit blocks is expressed by:

$$P = NE/PE,$$

where P is the power value of each window, NE is a number of elements in a first region of the integrated circuit design, and PE is a power consumed by the circuit block or the heater within the first region.

5. The method of claim 3, wherein the determining the heat signature of the integrated circuit design from at least the configured power information or the location information for each circuit block of the set of circuit blocks or each heater of the set of heaters included in the integrated circuit design further comprises:

for each window, determining a total heat value (HT) of a window based on a power value of the window and a power value of other windows.

6. The method of claim 5, wherein the determining the total heat value (HT) of the window based on the power value of the window and the power value of other windows is expressed by:

$$HT = \sum_{j=1}^{m-1} \sum_{i=1}^{n-1} k * \frac{P_{ij}}{D_{ij}}$$

where P_{ij} is a power value of ith and jth indices, k is a constant based on a material of the circuit block or the heater, D_{ij} is a distance between the ith and jth window, m is a number of rows in the grid of windows, n is a number of columns in the grid of windows.

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7. The method of claim 5, wherein the determining the heat signature of the integrated circuit design from at least the configured power information or the location information for each circuit block of the set of circuit blocks or each heater of the set of heaters included in the integrated circuit design further comprises:

populating a heat map according to the total heat value of each window.

8. The method of claim 1, wherein the first heat distribution throughout the integrated circuit design is uniform.

9. The method of claim 1, wherein the simultaneously performing the burn-in test of the integrated circuit and the automated test of the integrated circuit are performed without an oven or a burn-in board.

10. The method of claim 1, wherein performing the simulation of the first heat distribution throughout the integrated circuit design comprises:

modifying the integrated circuit design based on the first heat distribution throughout the integrated circuit design.

11. A method of testing an integrated circuit on a test circuit board, the method comprising:

performing, by a processor, a simulation of a first heat distribution throughout an integrated circuit design, the integrated circuit design including at least a set of circuit blocks, the performing the simulation comprising:

obtaining configured power information for at least each circuit block of the set of circuit blocks in the integrated circuit design;

determining a heat signature of the integrated circuit design from at least the configured power information for each circuit block of the set of circuit blocks included in the integrated circuit design, the heat signature including heat values distributed throughout the integrated circuit design; and

modifying the integrated circuit design in response to at least determining that the heat values of the heat signature of the integrated circuit design are not within a heat range;

manufacturing the integrated circuit according to the integrated circuit design; and

simultaneously performing a burn-in test of the integrated circuit and an automated test of the integrated circuit, the integrated circuit being configured to operate according to simulated design power levels and being coupled to the test circuit board, wherein the burn-in test has a minimum burn-in temperature of the integrated circuit and a burn-in heat distribution across the integrated circuit.

12. The method of claim 11, further comprising:

not modifying the integrated circuit design in response to at least determining that the heat values of the heat signature of the integrated circuit design are within the heat range or determining that a number of iterations of the simulation exceeds a user defined limit.

13. The method of claim 11, wherein modifying the integrated circuit design comprises:

modifying a configured power of at least an element in the integrated circuit design in response to determining that the heat values of the heat signature of the integrated circuit design are not within the heat range of the integrated circuit design, the element including at least a first circuit block of the set of circuit blocks.

14. The method of claim 13, wherein modifying the configured power of at least the element in the integrated circuit design comprises:

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increasing the configured power of at least the first circuit block of the set of circuit blocks.

15. The method of claim 13, wherein modifying the configured power of at least the element in the integrated circuit design comprises:

decreasing the configured power of at least the first circuit block of the set of circuit blocks.

16. The method of claim 11, wherein performing the simulation of the first heat distribution throughout the integrated circuit design further comprises:

extracting location information for each circuit block of the set of circuit blocks in the integrated circuit design from a design file.

17. The method of claim 11, wherein the first heat distribution throughout the integrated circuit design is uniform.

18. The method of claim 11, wherein the simultaneously performing the burn-in test of the integrated circuit and the automated test of the integrated circuit are performed without an oven or a burn-in board.

19. A testing system comprising:

an integrated circuit;

a carrier wafer coupled to at least the integrated circuit; and

a first system electrically coupled to the integrated circuit, the first system comprising:

a non-transitory computer readable medium configured to store non-transitory instructions; and

a processor coupled to the non-transitory computer readable medium, wherein the processor is configured to execute the non-transitory instructions for:

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performing a simulation of a first heat distribution throughout an integrated circuit design, the integrated circuit design corresponding to the integrated circuit;

wherein the testing system is configured to simultaneously perform a burn-in test of the integrated circuit and an automated test of the integrated circuit, wherein the burn-in test has a minimum burn-in temperature of the integrated circuit and a burn-in heat distribution across the integrated circuit, wherein the testing system configured to simultaneously perform the burn-in test of the integrated circuit and the automated test of the integrated circuit comprises:

configuring a set of circuit blocks and a first set of heaters as a first set of heat sources for the burn-in test of the integrated circuit thereby generating a first heat signature of the integrated circuit.

20. The testing system of claim 19, wherein the testing system configured to simultaneously perform the burn-in test of the integrated circuit and the automated test of the integrated circuit further comprises:

placing the integrated circuit on the carrier wafer; and

configuring at least a portion of the carrier wafer as a second set of heat sources for the burn-in test of the integrated circuit thereby generating a second heat signature of the integrated circuit,

wherein the second set of heat sources corresponding to a second set of heaters positioned in a grid arrangement of an integrated circuit die, and the integrated circuit die is part of the carrier wafer.

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